

26020/26021 Chemistry (Polytechnical Foundation) Formula Collection

2026

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1 Symbols, Notation, and Units

1.1 Ctrl+F keywords: symbol, notation, units, constant, subscript, standard state, phase, concentration, activity

Use this index when a formula contains an unfamiliar quantity or mark. Topic sections repeat the symbols needed for their calculations.

1.2 Python toolbox reference

The calculation notebook is `exam_toolbox.ipynb`; it imports `exam_tools` from `support_material/input/exams`. Use it when a recipe below names a Python function, especially for formula parsing, nonlinear equilibrium roots, logarithmic/exponential relations, regression, or repeated unit conversions. The toolbox also exports physical constants; use `constant(...)` or `constant_info(...)` to check values such as R , F , N_A , h , and c .

1.3 Common calculation quantities

Table 1.1: Common calculation quantities and typical units.

Symbol	Meaning	Typical unit
n	Amount of substance; context can change this meaning as listed below	mol
m	Mass; in colligative-property formulas, italic m may instead mean molality	g, kg, or mol kg ⁻¹
M	Molar mass	g mol ⁻¹
c , [A]	Amount concentration of a species A in solution	mol L ⁻¹
V	Volume	L or m ³
p , P	Pressure or vapor pressure, according to context	bar, atm, or Pa
T	Absolute temperature	K
ρ	Density	g cm ⁻³ or kg m ⁻³
R	Gas constant, used with units matching the formula data	e.g. J mol ⁻¹ K ⁻¹
N_A	Avogadro constant, 6.022×10^{23} particles per mole	mol ⁻¹
ν_i	Stoichiometric coefficient of species i	dimensionless
ξ	Reaction extent	mol

1.4 Thermodynamics, equilibrium, kinetics, and electrochemistry

Table 1.2: Thermodynamics, equilibrium, kinetics, and electrochemistry symbols.

Symbol	Meaning	Typical unit
q , w	Heat transferred to the system; work done on the system	J or kJ
U , H , S , G	Internal energy, enthalpy, entropy, Gibbs free energy	J, kJ, or J K ⁻¹
Δ	Change: final value minus initial value, or reaction change	as quantity
Q	Reaction quotient computed from current composition	dimensionless
K	Equilibrium constant; subscripts identify the equilibrium type	dimensionless
r	Reaction rate	mol L ⁻¹ s ⁻¹
k	Rate constant; units depend on reaction order	order-dependent
E_a	Activation energy	J mol ⁻¹
E_{cell}	Electrochemical cell potential	V
F	Faraday constant, charge per mole of electrons	C mol ⁻¹

1.5 Marks, subscripts, and chemical notation

Table 1.3: Marks, subscripts, phases, and chemical notation.

Notation	Meaning
$^\circ$	Standard-state value, normally referring to stated standard conditions
rxn, f, comb	Reaction, formation, and combustion quantity
vap, fus, sub	Vaporization, fusion, and sublimation
tot, ext, 0	Total, external, and initial quantities
sp, cell, eq	Solubility product, electrochemical cell, and equilibrium
a_A	Activity of species A; dilute-solution calculations often use concentration in its place
(s), (l), (g), (aq)	Solid, liquid, gas, and aqueous phase
$\longrightarrow$, $\rightleftharpoons$	Reaction arrow and reversible/equilibrium arrow
ln, log	Natural logarithm and base-10 logarithm
Σ , $\sum$	Sum over the indicated species or terms

1.6 Symbols with multiple meanings

Table 1.4: Symbols whose meaning must be identified from context.

Symbol	Meaning determined by context
n	Amount in moles in stoichiometry/gases; principal quantum number in electronic structure; moles of electrons in electrochemistry; degree of polymerization in polymers.
M	Molar mass in calculations; the generic metal symbol in dissolution formulas such as MX_2 .
Q , q	Capital Q is reaction quotient; lowercase q is heat.
p , P	Lowercase p is generally gas/partial pressure; uppercase P is used for vapor pressure in the phase section.

2 Atoms, Stoichiometry, and Nomenclature

2.1 Ctrl+F keywords: isotope, neutron, ion formula, naming, molar mass, empirical formula, limiting reactant, yield, navngivning, stofmaengde, udbytte

Use for isotope particles, compound formulas and names, composition, sum formulas, reaction amounts, limiting reagent, theoretical yield, and percent yield.

2.2 Symbols and notation

Table 2.1: Symbols used for atoms, composition, and stoichiometry

Symbol	Meaning and usage
A , Z , N_n	Mass number, atomic number/proton count, and neutron count.
X	Element symbol in isotope notation ${}_Z^AX$.
N , N_A	Number of particles and Avogadro constant.
n , m , M	Amount in moles, mass, and molar mass.
w_i	Mass percent of component i .
ν_i , ξ	Balanced-reaction coefficient and reaction extent.
k	Integer factor from an empirical formula to a molecular formula.
q_i , a_i	Charge and formula subscript of ion i .

2.3 Core formulas and definitions

2.3.1 Isotope notation, mass number, atomic number, neutron count

$${}_Z^AX, \quad A = Z + N_n, \quad N_n = A - Z \quad (2.1)$$

Z is the proton count. A neutral atom has Z electrons; an ion of charge $r+$ has $Z - r$ electrons and an ion of charge $r-$ has $Z + r$ electrons.

2.3.2 Mole, particles, and molar mass

$$N = nN_A, \quad N_A = 6.022 \times 10^{23} \text{ mol}^{-1}, \quad n = \frac{m}{M}, \quad m = nM \quad (2.2)$$

M is the sum of atomic molar masses multiplied by formula subscripts.

2.3.3 Mass percent and elemental composition

$$w_i = \frac{m_i}{m_{\text{total}}} \times 100\%, \quad w_{\text{element}} = \frac{a_i M_i}{M_{\text{compound}}} \times 100\% \quad (2.3)$$

In the second expression, a_i is the number of atoms of the element per formula unit.

2.3.4 Empirical formula and molecular formula

$$n_i = \frac{m_i}{M_i}, \quad r_i = \frac{n_i}{\min(n_i)} \quad (2.4)$$

$$k = \frac{M_{\text{molecular}}}{M_{\text{empirical}}}, \quad \text{molecular formula} = k (\text{empirical formula}) \quad (2.5)$$

Accept k only when it is approximately an integer; convert fractional r_i values to the smallest whole-number ratio.

2.3.5 Balanced equation stoichiometry and limiting reactant

$$\nu_A A + \nu_B B \rightarrow \nu_C C, \quad n_C = n_A \frac{\nu_C}{\nu_A} \quad (2.6)$$

$$\xi_i = \frac{n_i}{\nu_i}, \quad \xi_{\max} = \min(\xi_i), \quad n_{\text{product}} = \xi_{\max} \nu_{\text{product}} \quad (2.7)$$

$$\% \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\% \quad (2.8)$$

Coefficients are mole ratios. Balance atoms and charge before applying them.

2.4 Ionic formulas and names

2.4.1 Charge neutrality for ionic compounds

$$\sum_i q_i a_i = 0 \quad (2.9)$$

Choose the smallest whole-number subscripts that satisfy (2.9). Use parentheses when a polyatomic ion has a subscript larger than one, for example $(\text{NH}_4)_2\text{HPO}_4$.

2.4.2 Common ions, English and Danish search names

Table 2.2: Common polyatomic ions for formula writing and naming

Ion	Name / search alias	Ion	Name / search alias
NH_4^+	ammonium	OH^-	hydroxide
NO_3^-	nitrate	NO_2^-	nitrite
SO_4^{2-}	sulfate / sulfat	SO_3^{2-}	sulfite / sulfit
HSO_4^-	hydrogen sulfate	CO_3^{2-}	carbonate / carbonat
PO_4^{3-}	phosphate / fosfat	HPO_4^{2-}	hydrogen phosphate
H_2PO_4^-	dihydrogen phosphate	CN^-	cyanide

2.4.3 Binary molecular prefixes

Table 2.3: Prefixes for binary molecular compounds

1	mono	2	di	3	tri
4	tetra	5	penta	6	hexa
7	hepta	8	octa	9	nona
10	deca				

For binary molecular compounds, the second element receives the ending “-ide”;
omit “mono” on the first element in common naming.

2.5 Exam recipe: isotope particle count

Use when: isotope notation, mass number, atomic number, neutrons, protons, or electrons in an ion are requested.

Given / Find: Given ${}^A_Z\text{X}$ and possibly an ionic charge; find proton, neutron, and electron counts.

Required references: Equation (2.1) and notation in Table 2.1.

Steps:

1. Read the lower index as Z , so the proton count is Z , using (2.1).
2. Compute neutrons as $N_n = A - Z$ using (2.1).
3. Set electrons equal to Z for a neutral atom; for a charged ion subtract the positive charge or add the magnitude of a negative charge.

Checks: The neutron count is nonnegative; isotopes of one element have identical Z .

Common traps: Confusing mass number with molar mass; adding electrons for a positive ion.

2.6 Exam recipe: ionic formula and naming

Use when: ions, charges, or an ionic compound name must be converted to a neutral formula or back to a name.

Given / Find: Given cation and anion names or charges; find the smallest neutral formula and systematic name.

Required references: Charge-neutrality equation (2.9) and common-ion Table 2.2.

Steps:

1. Identify each ion and its charge from the problem or Table 2.2.
2. Choose the smallest subscripts whose signed charge sum is zero by applying (2.9).
3. Write the cation first and place parentheses around a repeated polyatomic ion.
4. Name the cation followed by the anion; state the oxidation number for a metal with variable charge when needed to identify q_i .

Checks: Substitute subscripts and charges into (2.9); the result must be zero.

Common traps: Leaving reducible subscripts; changing the internal subscript of a polyatomic ion; using molecular prefixes for ionic compounds.

2.7 Exam recipe: molar mass and mass percent

Use when: formula mass, molar mass, elemental mass percentage, or composition is requested.

Given / Find: Given a chemical formula and atomic molar masses; find M_{compound} or w_{element} .

Required references: Equations (2.2) and (2.3).

Python: Brug `exam_toolbox.ipynb` med `molar_mass("Ca(OH)2")` for formulaer med parenteser, hydrater eller gentagne molarmasser; brug resultatet direkte i masseprocentformlen.

Steps:

1. Count each element's atoms from the formula, multiplying through any parentheses.
2. Compute M_{compound} by summing each atom count times its atomic molar mass as specified below (2.2).
3. For an element percentage, divide its atom-count contribution $a_i M_i$ by M_{compound} and multiply by 100% using (2.3).

Checks: Element percentages in a complete composition sum to approximately 100%; molar mass units are g mol^{-1} .

Common traps: Ignoring parentheses or hydration coefficients; dividing an atomic mass by the compound mass without its subscript.

2.8 Exam recipe: empirical/molecular formula

Use when: masses or percentages of elements and possibly a compound molar mass are supplied.

Given / Find: Given elemental masses or mass percentages; find empirical formula and, if $M_{\text{molecular}}$ is given, molecular formula.

Required references: Equations (2.4), (2.5), and (2.2).

Python: Brug `exam_toolbox.ipynb` med `molar_mass(...)` til at kontrollere empirisk-formel- og molekylformel-masser, naar formelen er fundet.

Steps:

1. For percentages, choose a 100 g basis so each percentage becomes an element mass.
2. Convert each element mass to moles using $n = m/M$ in (2.2).
3. Divide each mole amount by the smallest using (2.4); multiply all ratios by one common small integer if they are near simple fractions.
4. Write the empirical formula from the resulting whole-number subscripts and compute its molar mass.
5. If a molecular molar mass is supplied, calculate k using (2.5) and multiply every empirical subscript by that integer.

Checks: Reconstructed percentages agree with the data; k is approximately a positive integer.

Common traps: Rounding 1.50 to 2 instead of scaling all ratios; scaling only one subscript.

2.9 Exam recipe: stoichiometry/limiting reactant/yield

Use when: a balanced reaction connects reactant masses or moles to product amount, excess reagent, theoretical yield, or percent yield.

Given / Find: Given a reaction and one or more amounts; find limiting reactant, product quantity, or yield.

Required references: Equations (2.2), (2.6), (2.7), and (2.8).

Python: Brug `exam_toolbox.ipynb` med `molar_mass(...)` for alle masse-til-mol og mol-til-masse trin i stoikiometri med sammensatte formler.

Steps:

1. Balance the chemical reaction so the coefficients ν_i required by (2.6) conserve atoms and charge.
2. Convert each available reactant quantity to moles using (2.2).
3. Compute each available extent n_i/ν_i and select the smallest value as ξ_{max} using (2.7).
4. Compute product moles as $\xi_{\text{max}}\nu_{\text{product}}$ from (2.7), then convert to the requested mass or particle count with (2.2).
5. If actual product is stated, divide it by the theoretical product in identical units using (2.8).

Checks: The limiting reagent is consumed first; percent yield is normally at most 100%; units cancel correctly.

Common traps: Using grams as coefficient ratios; choosing the largest n_i/ν_i ; comparing actual mass with theoretical moles.

3 Aqueous Reactions, Redox Balancing, and Gases

3.1 Ctrl+F keywords: molarity, dilution, net ionic, precipitation, redox, oxidation number, gas law, partial pressure, ideal gas, gas stoichiometry, pressure tank, bundfald

Use for solution concentrations, net ionic equations, precipitate decisions, half-reaction balancing, gas state changes, gas mixtures, and reaction amounts involving gases.

3.2 Symbols and notation

Table 3.1: Symbols used for aqueous reactions, redox, and gases

Symbol	Meaning and usage
c, n, V	Molar concentration, amount of substance, and solution or gas volume.
c_1, V_1, c_2, V_2	Concentration and volume before and after dilution.
$Q_{\text{sp}}, K_{\text{sp}}$	Ion product at current concentrations and solubility-product constant.
p, T, R	Gas pressure, absolute temperature, and gas constant.
x_i, p_i, p_{tot}	Mole fraction, partial pressure of gas i , and total pressure.
ρ, M	Gas density and molar mass.
e^-, H^+, OH^-	Species used in half-reaction balancing.
(aq), (s)	Dissolved aqueous species and solid precipitate.

3.3 Solutions and aqueous reactions

3.3.1 Molarity, dilution, actual ion concentration

$$c = \frac{n}{V}, \quad n = cV, \quad c_1 V_1 = c_2 V_2 \quad (3.1)$$

Use liters for V . For a completely dissociated strong electrolyte, obtain an ion concentration by multiplying c by its coefficient; for example, $\text{CaCl}_2(\text{aq}) \longrightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$ gives $[\text{Cl}^-] = 2c$.

3.3.2 Complete dissolution and total ion amount



For a strong soluble electrolyte, each formula unit produces the number of dissolved ions shown by the balanced dissociation coefficients.

3.4 Exam recipe: moles of ions from complete dissolution

Use when: a soluble ionic compound dissolves completely and the task asks for total moles of ions or the amount of one ion.

Given / Find: Given the amount or mass of a soluble salt; find $n_{\text{ions, total}}$ or n of a specified ion.

Required references: Equations (3.2) and (3.1); use Equation (2.2) if a mass is given.

Python: Brug `exam_toolbox.ipynb` med `molar_mass(...)` naar den givne saltmasse foerst skal omregnes til stofmaengde.

Steps:

1. Write the complete dissociation equation and balance the ionic coefficients; for example, one $\text{Al}_2(\text{SO}_4)_3$ formula unit produces $2 + 3 = 5$ ions.
2. Convert a stated mass or solution concentration to moles of salt before applying the dissociation coefficients.
3. Multiply salt moles by the requested ion coefficient, or by the sum of all ion coefficients for total moles of ions, using (3.2).

Checks: Total charge of ions produced from one formula unit is zero; total ion moles exceed salt moles when more than one ion is formed.

Common traps: counting distinct ion types instead of stoichiometric ion amounts; forgetting parentheses in salts such as $\text{Al}_2(\text{SO}_4)_3$.

3.4.1 Net ionic equation and spectator ions



Remove ions that occur unchanged on both sides. A net ionic equation conserves atoms and total charge.

3.4.2 Precipitation recognition

$$Q_{\text{sp}} > K_{\text{sp}} \Rightarrow \text{precipitate}, \quad Q_{\text{sp}} < K_{\text{sp}} \Rightarrow \text{no precipitate yet} \quad (3.4)$$

Table 3.2: Recognition cues for precipitation questions

Problem wording	Immediate action
mixing ionic solutions; bundfald	Form candidate products and test whether one is insoluble.
K_{sp} or sparingly soluble	Compute post-mixing ion concentrations and compare Q_{sp} with K_{sp} .
net ionic equation	Dissociate soluble strong electrolytes and cancel spectator ions.

Table 3.3: Common solubility rules for deciding candidate precipitates when no K_{sp} data are supplied

Ions in compound	Rapid solubility decision
NO_3^- , group 1 cations, NH_4^+	Treat as soluble; these usually remain spectator ions.
Cl^- , Br^- , I^-	Usually soluble; treat compounds with Ag^+ or Pb^{2+} as precipitate candidates.
SO_4^{2-}	Usually soluble; treat compounds with Ba^{2+} or Pb^{2+} as precipitate candidates.
CO_3^{2-} , PO_4^{3-} , OH^-	Usually insoluble unless paired with group 1 cations or NH_4^+ .

3.5 Exam recipe: net ionic precipitation

Use when: two aqueous ionic solutions are mixed and the question asks for a precipitate or net ionic equation.

Given / Find: Given dissolved compounds, concentrations, and possibly K_{sp} ; find whether a solid forms and write its net ionic equation.

Required references: Equations (3.1), (3.3), (3.4), and Tables 3.2 and 3.3.

Steps:

1. Dissociate each soluble strong electrolyte into ions and obtain concentrations after mixing with $c = n/V$ from (3.1) when numerical data are given.
2. Pair the available cations and anions to write candidate neutral solid formulas by charge balance.
3. Decide the precipitate branch: when no numerical solubility data are supplied, classify each candidate using Table 3.3; when K_{sp} is supplied, calculate Q_{sp} from mixed ion concentrations and apply (3.4).
4. Write only the precipitating ions and the solid product, following the pattern in (3.3), then choose coefficients that conserve atoms and charge.

Checks: No unchanged spectator ion remains; phases and charges are shown; both atom count and charge balance.

Common traps: Using concentrations before dilution by mixing; leaving nitrate or alkali metal ions in the net equation; writing an uncharged ionic product with incorrect subscripts.

3.5.1 Gravimetric concentration from a precipitate

$$n_{\text{precipitate}} = \frac{m_{\text{precipitate}}}{M_{\text{precipitate}}}, \quad n_{\text{analyte}} = n_{\text{precipitate}} \frac{\nu_{\text{analyte}}}{\nu_{\text{precipitate}}}, \quad c_{\text{analyte}} = \frac{n_{\text{analyte}}}{V_{\text{sample}}} \quad (3.5)$$

The coefficient ratio is taken from the balanced net ionic precipitation reaction.

3.6 Exam recipe: concentration from precipitate mass

Use when: an ion is precipitated quantitatively, the mass of solid is measured, and the original solution concentration is requested.

Given / Find: Given a precipitate mass, its identity, the sampled solution volume, and the analyte ion; find the analyte concentration.

Required references: Equation (3.5), Equation (2.2), and the net ionic reaction method in Equation (3.3).

Python: Brug `exam_toolbox.ipynb` med `molar_mass(...)` for praecipitatets molarmasse, især ved salte med par-entesser eller tunge ioner.

Steps:

1. Write the balanced net ionic reaction linking the analyte ion to the measured solid; its coefficients define the mole ratio.
2. Calculate the precipitate molar mass, then convert measured solid mass to moles using the first expression in Equation (3.5).
3. Convert precipitate moles to analyte moles using the coefficient ratio in the second expression of Equation (3.5).
4. Divide analyte moles by the original sampled solution volume in liters using the final expression of Equation (3.5).

Checks: For $\text{Ag}^+ + \text{Cl}^- \longrightarrow \text{AgCl(s)}$, precipitate and chloride moles are equal; concentration units are mol L^{-1} .

Common traps: dividing by total reagent-plus-sample volume rather than the original sample volume; using mass ratios in place of the balanced mole ratio.

3.7 Redox balancing

3.7.1 Oxidation, reduction, and electron cancellation

Oxidation produces electrons and increases oxidation state; reduction consumes electrons and decreases oxidation state. The final equation contains no free electrons.

3.7.2 Half-reaction sequence in acid and base

$$\begin{aligned} &\text{balance atoms except H and O; balance O with H}_2\text{O;} \\ &\text{balance H with H}^+; \text{ balance charge with e}^-; \\ &\text{in base, add OH}^- \text{ to both sides for every H}^+ \text{ and cancel H}_2\text{O.} \end{aligned} \tag{3.6}$$

3.7.3 Oxidation-state formula:

$$\sum_i n_i \text{OS}_i = q_{\text{species}} \tag{3.7}$$

where n_i is the number of atoms of element i , OS_i is its oxidation state, and q_{species} is the total charge of the molecule or ion. Assign known values first, usually $\text{O} = -II$, $\text{H} = +I$, free elements = 0, and monatomic ions = their ion charge; set the unknown oxidation state to x and solve.

3.8 Exam recipe: redox balancing

Use when: an oxidation–reduction equation must be balanced in acidic or basic aqueous solution.

Given / Find: Given reactant and product species plus acidic or basic medium; find the balanced overall equation.

Required references: Oxidation-state rules in Table 11.2; half-reaction procedure (3.6).

Steps:

1. Identify oxidation states with (3.7) for the atoms that change: multiply each atom count by its oxidation state, set the sum equal to the species charge, and solve any unknown x .
2. Separate oxidation and reduction half-reactions: oxidation state increases for oxidation and decreases for reduction.
3. For each half-reaction, balance non-H, non-O atoms, then add H_2O , H^+ , and e^- in the order prescribed by (3.6).
4. Multiply each balanced half-reaction by the smallest integers that give equal electron counts, then add and cancel identical species.
5. For basic solution, apply the final conversion step in (3.6) and cancel excess water.

Checks: Total atoms and total charge agree on both sides; electrons do not occur in the final equation; no H^+ remains in a final basic-medium equation.

Common traps: Balancing charge before oxygen and hydrogen; converting to base before the acidic half-reaction is balanced; failing to cancel water.

3.9 Gases

3.9.1 Ideal gas law and fixed-amount state change

$$pV = nRT \quad (3.8)$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \quad (n \text{ constant}) \quad (3.9)$$

Temperatures must be in kelvin. Choose R to match the pressure and volume units.

3.9.2 Partial pressure and mole fraction

$$p_i = x_i p_{\text{tot}}, \quad x_i = \frac{n_i}{n_{\text{tot}}}, \quad p_{\text{tot}} = \sum_i p_i \quad (3.10)$$

For an ideal gas mixture, mole fraction equals pressure fraction.

3.9.3 Gas density and molar mass

$$\rho = \frac{m}{V} = \frac{pM}{RT}, \quad M = \frac{\rho RT}{p} \quad (3.11)$$

Use (3.11) when density, pressure, and temperature identify an unknown gas.

3.10 Exam recipe: molar mass and molecular formula from gas data

Use when: an unknown gaseous compound is specified by mass and gas-state data and its molar mass or molecular formula is requested.

Given / Find: Given gas mass, pressure, volume, temperature, and optionally an empirical formula; find M and/or the molecular formula.

Required references: Equations (3.8) and (3.11); use Equation (2.5) when an empirical formula is given.

Python: Brug `exam_toolbox.ipynb` med `ideal_gas(...)` til at finde gasmol eller den manglende p, V, n, T , og `molar_mass(...)` til empirisk- og molekylformelmassen.

Steps:

1. Convert temperature to kelvin and use consistent pressure-volume units with the chosen gas constant.
2. Calculate gas moles as $n = pV/(RT)$ from Equation (3.8), then calculate $M = m/n$; equivalently calculate $\rho = m/V$ and use Equation (3.11).
3. If an empirical formula is supplied or obtained, calculate its formula mass and determine the integer factor $k = M/M_{\text{empirical}}$ by Equation (2.5).
4. Multiply every empirical-formula subscript by the nearest justified integer k to state the molecular formula.

Checks: M must be positive and close to an integer multiple of the empirical-formula mass when a molecular formula is requested.

Common traps: using Celsius; treating gas volume as molar mass information without pressure and temperature; rounding k before checking measurement precision.

3.10.1 Effusion through a small opening, Graham's law

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \quad (3.12)$$

At the same temperature, the lighter gas effuses faster because its molar mass is smaller.

3.11 Exam recipe: gas law state change

Use when: a fixed amount of gas changes pressure, volume, or temperature without a chemical reaction, such as a sealed tank or piston problem.

Given / Find: Given two gas states with one unknown variable; find final pressure, volume, or temperature.

Required references: Combined gas law (3.9); use (3.8) only when moles must first be determined.

Python: Brug `exam_toolbox.ipynb` med `ideal_gas(...)` naar en af p, V, n, T mangler eller naar gas tilføjes, fjernes eller forbruges.

Steps:

1. Confirm that the amount of gas is constant; otherwise calculate or track n with (3.8) instead of applying (3.9).
2. Convert each temperature to kelvin and put pressure and volume in matching units on both sides.
3. Substitute the two states into (3.9) and algebraically isolate the requested variable.

Checks: At constant volume, increasing absolute temperature increases pressure; at constant temperature, increasing volume decreases pressure.

Common traps: Substituting Celsius; applying the combined gas law when gas is added, removed, or consumed.

3.12 Exam recipe: gas stoichiometry

Use when: a reacting gas is specified by pressure, volume, temperature, or partial pressure, including a closed heated tank in which gases react before a final pressure is requested.

Given / Find: Given a balanced reaction and gas-state data; find reacting moles, product mass, product gas volume, or final gas pressure.

Required references: Equations (3.8), (3.10), and (2.6).

Python: Brug `exam_toolbox.ipynb` med `ideal_gas(...)` til gasmol/tryk efter reaktion og `molar_mass(...)` til produkt- eller reaktantmasser.

Steps:

1. If the gas is in a mixture, calculate its partial pressure with (3.10); otherwise use the stated gas pressure.
2. Convert the reacting gas data to moles by rearranging (3.8) as $n = pV/(RT)$, using kelvin and compatible units.
3. Write and balance the reaction, then apply the coefficient ratio exactly as in (2.6).
4. For a final reacting-gas mixture, subtract consumed reactants, add gaseous products, sum remaining gas moles, and use (3.8) at the final T and V ; otherwise convert product moles directly to the requested mass, volume, or pressure.

Checks: The pressure used is the reacting gas pressure; balanced coefficients relate moles, not liters unless state conditions are identical.

Common traps: Using total pressure for one component; applying gas volume ratios at different temperatures or pressures; using Celsius in $pV = nRT$.

3.13 Exam recipe: gas effusion from a punctured container

Use when: a container is punctured, gases escape through a small hole, or the task compares which gas remains or escapes faster.

Given / Find: Given two gas identities or molar masses in one container; find relative effusion rate or enrichment of the remaining gas.

Required references: Equation (3.12).

Python: Brug `exam_toolbox.ipynb` med `molar_mass(...)` naar effusionssammenligningen kræver molarmasser fra kemiske formler.

Steps:

1. Determine each molar mass from its formula.
2. Compare rates using (3.12): the gas with smaller M has larger effusion rate.

3. For a punctured mixture, state that the faster-effusing gas becomes depleted more quickly in the container, so the remaining gas mixture becomes richer in the heavier gas.

Checks: If $M_1 < M_2$, then $r_1/r_2 > 1$; identical molar masses give equal rates.

Common traps: using molecular size intuition instead of molar mass; concluding that the faster gas becomes enriched inside the leaking container.

4 Thermodynamics and Thermochemistry

4.1 Ctrl+F keywords: enthalpy, Hess, formation enthalpy, combustion, entropy, Gibbs, spontaneity, equilibrium constant

Use for heat, work, ΔH , ΔS , ΔG , equilibrium constants from thermodynamics, and temperature-dependent spontaneity.

4.2 Symbols and notation

Table 4.1: Thermodynamic symbols and notation.

Symbol	Meaning and usage
U, H, S, G	Internal energy, enthalpy, entropy, and Gibbs free energy.
q, w	Heat entering the system and work performed on the system.
p, q_p	Pressure and heat measured at constant pressure.
$p_{\text{ext}}, \Delta V$	External pressure and volume change in pressure-volume work.
T, R	Absolute temperature and gas constant.
ν_i	Coefficient of species i in the balanced reaction.
$^\circ, \text{rxn}, \text{f}, \text{comb}$	Standard state; reaction, formation, and combustion subscripts.
$\text{prod}, \text{react}, k$	Products, reactants, and reaction-equation term index in summations.
K, Q	Equilibrium constant and current reaction quotient.
D	Average bond enthalpy used for bond-breaking and bond-forming estimates.

4.3 Core formulas

4.3.1 First law, heat, and work

$$\Delta U = q + w, \quad w = -p_{\text{ext}}\Delta V \quad (4.1)$$

Heat into the system is positive. Expansion work is negative for the system when $\Delta V > 0$.

4.3.2 Constant-pressure heat and enthalpy

$$H = U + pV, \quad q_p = \Delta H \quad (4.2)$$

At constant pressure, measured heat equals the enthalpy change. $\Delta H < 0$ is exothermic and $\Delta H > 0$ is endothermic.

4.3.3 Reaction enthalpy from standard formation enthalpies

$$\Delta H_{\text{rxn}}^\circ = \sum_{\text{prod}} \nu_i \Delta H_{\text{f},i}^\circ - \sum_{\text{react}} \nu_i \Delta H_{\text{f},i}^\circ \quad (4.3)$$

Use coefficients from the balanced equation. Elements in their standard state have $\Delta H_{\text{f}}^\circ = 0$.

4.3.4 Hess's law and thermochemical scaling

$$\Delta H_{\text{overall}}^\circ = \sum_k \Delta H_k^\circ, \quad 2 \text{ A} \longrightarrow 2 \text{ B} : \Delta H = 2\Delta H_{\text{A} \longrightarrow \text{B}} \quad (4.4)$$

Reversing a reaction changes the sign. Multiplying a reaction multiplies ΔH , ΔS , and ΔG .

4.3.5 Combustion enthalpy

$$\Delta H_{\text{comb}}^{\circ} = \sum \nu \Delta H_{\text{f}}^{\circ}(\text{CO}_2, \text{H}_2\text{O}, \dots) - \sum \nu \Delta H_{\text{f}}^{\circ}(\text{fuel}, \text{O}_2) \quad (4.5)$$

Balance complete combustion to CO_2 and H_2O first. $\Delta H_{\text{f}}^{\circ}(\text{O}_2(\text{g})) = 0$.

4.3.6 Reaction enthalpy from average bond enthalpies

$$\Delta H_{\text{rxn}} \approx \sum D(\text{bonds broken}) - \sum D(\text{bonds formed}) \quad (4.6)$$

$$\Delta H_{\text{rxn}} \approx - \sum \Delta H_{\text{bond}}(\text{bonds broken}) + \sum \Delta H_{\text{bond}}(\text{bonds formed}) \quad (4.7)$$

when tabulated bond-formation values are negative

Bond enthalpy data are usually positive bond-breaking magnitudes; if the supplied table instead prints negative bond-formation values, use Equation (4.7). Count bonds from the balanced molecular reaction before applying either formula.

4.3.7 Entropy change from standard molar entropies

$$\Delta S_{\text{rxn}}^{\circ} = \sum_{\text{prod}} \nu_i S_i^{\circ} - \sum_{\text{react}} \nu_i S_i^{\circ} \quad (4.8)$$

Use $\text{J mol}^{-1} \text{K}^{-1}$ for entropy and convert to kJ if combining with ΔH .

4.3.8 Gibbs free energy and spontaneity

$$\Delta G = \Delta H - T\Delta S \quad (4.9)$$

$\Delta G < 0$ is spontaneous, $\Delta G = 0$ is equilibrium, and $\Delta G > 0$ is nonspontaneous under the stated conditions.

4.3.9 Standard free energy and equilibrium constant

$$\Delta G^{\circ} = -RT \ln K, \quad K = e^{-\Delta G^{\circ}/RT} \quad (4.10)$$

$K > 1$ corresponds to $\Delta G^{\circ} < 0$. Use kelvin and consistent energy units.

4.3.10 Nonstandard Gibbs free energy

$$\Delta G = \Delta G^{\circ} + RT \ln Q \quad (4.11)$$

The reaction direction under current conditions is determined by ΔG , not by ΔG° alone.

4.3.11 Spontaneity from signs and temperature

Table 4.2: Spontaneity predicted from signs of ΔH and ΔS .

ΔH	ΔS	Spontaneity
–	+	All T
+	–	No T
–	–	Low T
+	+	High T

$$T_{\text{switch}} = \frac{\Delta H}{\Delta S} \quad (\Delta G = 0) \quad (4.12)$$

4.4 Exam recipe: reaction enthalpy and Hess's law

Use when: the target is $\Delta H_{\text{rxn}}^{\circ}$ and formation enthalpies, combustion enthalpies, or thermochemical equations are provided.

Given / Find: Given a balanced or balanceable reaction and thermochemical data; find $\Delta H_{\text{rxn}}^{\circ}$ for the target reaction as written.

Required references: Equation (4.3), Equation (4.4), and, for combustion data, Equation (4.5).

Steps:

1. Balance the target reaction and retain its requested direction and stoichiometric scale.

2. If ΔH_f° values are given, insert the balanced coefficients in Equation (4.3).
3. If reaction equations are given, reverse or scale each equation until species cancel to the target; change or scale each ΔH° by Equation (4.4).
4. Add the adjusted enthalpy terms and report the value per target reaction as written; if heat per gram is requested, divide the molar reaction enthalpy by the mass of the specified substance consumed according to the balanced reaction.

Checks: Standard-state elements contribute 0 in formation-data calculations; complete combustion is normally exothermic, so $\Delta H_{\text{comb}}^\circ < 0$.

Common traps: Omitting stoichiometric coefficients; reversing a reaction without reversing the sign; changing the target equation after calculating its enthalpy; reporting kJ mol^{-1} when the question asks for kJ g^{-1} .

4.5 Exam recipe: reaction enthalpy from bond enthalpies

Use when: the problem supplies bond enthalpies or bond energies for reactant and product bonds instead of formation enthalpies.

Given / Find: Given a molecular reaction and bond enthalpies; find an estimated ΔH_{rxn} .

Required references: Equations (4.6) and (4.7).

Steps:

1. Balance the molecular reaction and draw or read enough structure to count each bond type broken in reactants and formed in products.
2. If the data are positive dissociation energies D , multiply each broken- and formed-bond count by D , then subtract the formed total from the broken total using Equation (4.6).
3. If the data are printed as negative bond-formation enthalpies, reverse the sign for bonds broken and retain the printed sign for bonds formed using Equation (4.7).

Checks: Forming more or stronger bonds than are broken should tend to make ΔH negative; count all bonds after balancing.

Common traps: Adding positive formed-bond dissociation energies instead of subtracting them; applying the positive- D formula directly to negative bond-formation entries.

4.6 Exam recipe: Gibbs spontaneity from signs and temperature

Use when: the question gives signs or numerical values of ΔH and ΔS and asks when the process is spontaneous.

Given / Find: Given ΔH , ΔS , and possibly T ; find the sign of ΔG , the spontaneous temperature range, or the threshold temperature.

Required references: Equation (4.9), Table 4.2, and Equation (4.12).

Steps:

1. Convert ΔS to the same energy unit basis as ΔH , normally $\text{kJ mol}^{-1}\text{K}^{-1}$.
2. For sign-only questions, select the matching row in Table 4.2.
3. For a stated temperature, calculate ΔG with Equation (4.9) and classify by its sign.
4. If spontaneity can switch with temperature, calculate T_{switch} from Equation (4.12) and state the spontaneous side of the threshold.

Checks: A computed threshold temperature must be in kelvin and physically meaningful; $\Delta G = 0$ at the threshold.

Common traps: Mixing J and kJ; interpreting $\Delta G > 0$ as spontaneous; using a threshold for the sign combinations that never switch.

4.7 Exam recipe: equilibrium constant from thermodynamic data

Use when: the target is K and standard Gibbs data or standard enthalpy and entropy data are supplied.

Given / Find: Given T and either $\Delta G_{\text{rxn}}^\circ$ or data for $\Delta H_{\text{rxn}}^\circ$ and $\Delta S_{\text{rxn}}^\circ$; find the dimensionless equilibrium constant K .

Required references: Equation (4.9), Equation (4.8), Equation (4.3), and Equation (4.10).

Python: Brug `exam_toolbox.ipynb` med `equilibrium_constant(...)` efter ΔG° er fundet; brug `constant("R")` hvis du regner eksponenten manuelt.

Steps:

1. Balance the target reaction and calculate any missing $\Delta H_{\text{rxn}}^\circ$ or $\Delta S_{\text{rxn}}^\circ$ using the balanced coefficients.
2. If $\Delta G_{\text{rxn}}^\circ$ is not given, calculate it at the stated T from Equation (4.9).
3. Convert $\Delta G_{\text{rxn}}^\circ$ to J mol^{-1} when using R in $\text{J mol}^{-1}\text{K}^{-1}$, and convert T to kelvin.
4. Calculate $K = e^{-\Delta G^\circ/RT}$ from Equation (4.10).

Checks: $\Delta G^\circ < 0$ must give $K > 1$, and $\Delta G^\circ > 0$ must give $K < 1$.

Common traps: Using Celsius; missing the minus sign in the exponent; mixing energy units; using Q when the request is the standard-state equilibrium constant.

5 Electronic Structure and Periodic Table

5.1 Ctrl+F keywords: quantum number, electron configuration, ion configuration, orbital, ionization energy, atomic radius, ionic radius, electronegativity

Use for questions about electron configurations, allowed quantum numbers, periodic trends, isoelectronic ions, and light-energy formulas.

5.2 Symbols and notation

Table 5.1: Electronic-structure symbols and notation.

Symbol	Meaning and usage
E, h, ν, λ, c	Photon energy, Planck constant, frequency, wavelength, and speed of light.
E_n, R_{H}	Energy of Bohr level n and the hydrogen energy constant used in that formula.
n_i, n_f	Initial and final principal quantum number in a transition.
p, m, v	Particle momentum, mass, and speed in the de Broglie relation.
n, ℓ, m_ℓ, m_s	Principal, angular-momentum, magnetic, and spin quantum numbers.
$\#e^-, \#\text{orbitals}$	Number of electrons and number of orbitals in a subshell.
$Z, \chi, \text{IE}, \text{EA}$	Atomic number, electronegativity, ionization energy, and electron affinity.

5.3 Light and quantization

5.3.1 Photon energy, wavelength, frequency

$$E = h\nu = \frac{hc}{\lambda} \quad (5.1)$$

Shorter wavelength means higher photon energy. Use SI units unless the data table specifies otherwise.

Python: Brug `exam_toolbox.ipynb` med `constant("h")` og `constant("c")` til fotonberegninger; `python.py` viser et kort SymPy-eksempel.

5.3.2 Bohr energy levels and transitions

$$E_n = -\frac{R_{\text{H}}}{n^2}, \quad \Delta E = E_{n_f} - E_{n_i} \quad (5.2)$$

Emission releases energy when the electron moves to lower n . Absorption requires positive energy input.

5.3.3 De Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad (5.3)$$

Use for matter-wave questions involving particle mass and speed.

5.4 Quantum numbers and orbitals

5.4.1 Allowed quantum numbers

Table 5.2: Allowed quantum numbers and meanings.

Symbol	Allowed values	Meaning
n	$1, 2, 3, \dots$	Shell
ℓ	$0, 1, \dots, n-1$	Subshell
m_ℓ	$-\ell, \dots, +\ell$	Orbital
m_s	$\pm \frac{1}{2}$	Spin

Invalid sets usually break one allowed-value rule in Table 5.2.

5.4.2 Subshell labels and capacities

$$\ell = 0, 1, 2, 3 \leftrightarrow s, p, d, f, \quad \# \text{orbitals} = 2\ell + 1, \quad \#e^- = 2(2\ell + 1) \quad (5.4)$$

Thus s, p, d, f hold 2, 6, 10, 14 electrons.

5.4.3 Electron configuration filling order

$$1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, \dots \quad (5.5)$$

Fill lower energy orbitals first, obey Pauli exclusion, and maximize unpaired spins within degenerate orbitals.

5.4.4 Electron count for atoms and ions

$$N_e = Z - q \quad (5.6)$$

Use signed charge q : a positive charge removes electrons and a negative charge adds electrons.

5.5 Periodic trends

5.5.1 Atomic radius trend

$$\text{Across a period: radius decreases,} \quad \text{down a group: radius increases} \quad (5.7)$$

Increasing effective nuclear charge contracts atoms across a period. New shells dominate down a group.

5.5.2 Ionization energy trend

$$\text{X(g)} \longrightarrow \text{X}^+(\text{g}) + \text{e}^-, \quad \text{IE} > 0 \quad (5.8)$$

Ionization energy generally increases across a period and decreases down a group. Filled and half-filled subshells can create exceptions.

5.5.3 Electron affinity and electronegativity

$$\text{X(g)}^+\text{e}^- \longrightarrow \text{X}^-(\text{g}), \quad \chi \approx \frac{\text{IE} + \text{EA}}{2} \quad (5.9)$$

Electronegativity increases toward fluorine. Use it to judge bond polarity and likely formal-charge placement.

5.5.4 Isoelectronic ionic radius

$$\text{Same electron count: larger } Z \Rightarrow \text{smaller radius} \quad (5.10)$$

For N^{3-} , O^{2-} , F^- , Na^+ , Mg^{2+} , the highest nuclear charge is smallest.

5.6 Exam recipe: quantum-number validity

Use when: the problem asks whether (n, ℓ, m_ℓ, m_s) is allowed or how many orbitals or electrons belong to a subshell.

Given / Find: Given one or more quantum-number sets or a specified subshell; find valid/invalid sets, the failed rule, or the capacity.

Required references: Table 5.2 and Equation (5.4).

Steps:

1. Check n against the first row of Table 5.2: it must be a positive integer.
2. Check ℓ against the chosen n : require $0 \leq \ell \leq n - 1$.
3. Check m_ℓ and m_s against their table rows: require $-\ell \leq m_\ell \leq +\ell$ and $m_s = \pm \frac{1}{2}$.
4. If capacity is requested, insert ℓ in Equation (5.4).

Checks: An s subshell must have one orbital and capacity two electrons; a proposed m_ℓ cannot have magnitude greater than ℓ .

Common traps: Allowing $\ell = n$; accepting $m_s = 0$; confusing the number of orbitals with the number of electrons.

5.7 Exam recipe: electron configuration and ion configuration

Use when: the problem asks for an atom or ion configuration, valence or unpaired electrons, or identification of an isoelectronic species.

Given / Find: Given an element identity or Z and an ionic charge; find its electron configuration or compare electron counts.

Required references: Equations (5.6), (5.5), and (5.4).

Steps:

1. Calculate the electron count from Equation (5.6), using $q = 0$ for a neutral atom and the signed ionic charge otherwise.
2. Add electrons in the orbital order of Equation (5.5), never exceeding each subshell capacity from Equation (5.4); inside a p , d , or f subshell, place one electron in each equal-energy orbital before pairing.
3. For a cation, remove electrons from the highest principal shell first; for transition-metal cations, remove ns before $(n - 1)d$.
4. Sum all superscripts in the completed configuration and verify that the total equals N_e ; when unpaired electrons are requested, count singly occupied orbitals after the Hund-rule placement in step 2.

Checks: A positive charge reduces the electron count; species identified as isoelectronic must have equal total superscripts; a completely filled subshell has no unpaired electrons.

Common traps: Subtracting electrons for an anion; removing $3d$ before $4s$ in transition-metal cations; exceeding a subshell capacity.

5.8 Exam recipe: ranking atomic or ionic radius and ionization energy

Use when: the problem asks for largest/smallest radius or highest/lowest first ionization energy among atoms or ions.

Given / Find: Given a list of species; find an ordered ranking of radius or ionization energy with the controlling rule stated.

Required references: Equation (5.7), Equation (5.10), and Equation (5.8).

Steps:

1. For ions, calculate electron counts; if counts match, rank radius using Equation (5.10).
2. For non-isoelectronic radii, apply Equation (5.7); for the same element, rank cation $<$ neutral atom $<$ anion in radius.
3. For first ionization energy, rank neutral atoms generally upward and to the right in the periodic table, then flag adjacent filled or half-filled subshell exceptions if relevant.
4. Write the final order with $<$ or $>$ defined explicitly as radius or ionization energy.

Checks: Within an isoelectronic series, the largest Z must give the smallest radius; a cation must not be ranked larger than its neutral parent atom.

Common traps: Applying neutral-atom trends directly to an isoelectronic ionic series; reversing a requested largest-to-smallest order; treating the general ionization-energy trend as exception-free.

6 Bonding, Lewis Structures, Formal Charge, and VSEPR

6.1 Ctrl+F keywords: Lewis, formal charge, resonance, VSEPR, bond angle, planar, polarity, dipole, hybridization

Use for drawing or checking Lewis structures, choosing resonance forms, molecular geometry, bond angles, polarity, and planar/non-planar questions.

6.2 Symbols and notation

Table 6.1: Bonding, Lewis, and VSEPR symbols.

Symbol	Meaning and usage
N_{val}, q	Total number of valence electrons and overall ionic charge.
FC, V	Formal charge and valence-electron count of the selected atom.
$N_{\text{nonbonding}}, N_{\text{bonding}}$	Nonbonding and bonding electrons assigned in a Lewis structure.
SN	Steric number: bonded atoms plus lone pairs on the central atom.
$\vec{\mu}_i, \vec{\mu}_{\text{mol}}, i$	Individual bond dipole, vector sum for molecular dipole, and bond index in the sum.
$\chi, \Delta\chi$	Electronegativity and electronegativity difference.
$N_{\text{bonding}}, N_{\text{antibonding}}$	Electrons in bonding and antibonding molecular orbitals in the bond-order formula.

6.3 Lewis structures

6.3.1 Valence electron count

$$N_{\text{val}} = \sum_i N_{\text{val},i} - q \quad (6.1)$$

For an ion, subtract positive charge and add negative charge. Total Lewis electrons must equal N_{val} .

6.3.2 Formal charge

$$\text{FC} = V - \left(N_{\text{nonbonding}} + \frac{1}{2} N_{\text{bonding}} \right) \quad (6.2)$$

Preferred structures usually minimize formal charges and place negative formal charge on more electronegative atoms.

6.3.3 Resonance structures

$$\text{resonance forms differ in electron placement, not atom placement} \quad (6.3)$$

Equivalent resonance forms indicate delocalization and equalized bond lengths. Do not move nuclei when drawing resonance.

6.3.4 Octet, expanded octet, electron-deficient exceptions

$$\text{period 2 atoms normally obey octet,} \quad \text{period 3 + may expand octet} \quad (6.4)$$

Hydrogen has a duet. Boron can be electron deficient; sulfur, phosphorus, iodine, and xenon can appear hypervalent in course examples.

6.4 Exam recipe: Lewis structure and formal charge

Use when: the problem asks for a valid Lewis structure, resonance contributors, formal charges, nitrate, or oxalate.

Given / Find: atom identities and charge q ; find electron placement, resonance forms, and formal charges.

Required references: Equation 6.1, Equation 6.2, Equation 6.3, and Equation 6.4.

Steps:

1. Compute N_{val} from Equation 6.1; reserve exactly that many electrons.
2. Choose a skeleton: hydrogen is terminal; otherwise put the least electronegative suitable atom centrally.
3. Draw single bonds, complete terminal-atom octets or hydrogen duets, then place remaining electrons centrally.
4. If a second-period central atom is electron deficient, convert adjacent lone pairs to multiple bonds without exceeding its octet.
5. Calculate FC on every atom using Equation 6.2; verify that the sum is q .
6. Draw equivalent electron-placement alternatives as resonance forms and select contributors with minimized, chemically sensible formal charges.

Checks: electrons drawn equal N_{val} ; formal charges sum to q ; no second-period atom exceeds an octet.

Common traps: moving atoms between resonance forms; ignoring ionic charge in the count; expanding a second-period octet.

6.5 VSEPR geometry

6.5.1 Steric number and electron domains

$$\text{SN} = \# \text{bonded atoms} + \# \text{lone pairs on central atom} \quad (6.5)$$

Multiple bonds count as one electron domain in VSEPR. Lone pairs compress bond angles.

6.5.2 Common VSEPR geometries and bond angles

Table 6.2: Common central-atom VSEPR geometries and ideal or approximate bond angles.

SN	Lone pairs	Molecular geometry	Bond angle
2	0	linear	180°
3	0	trigonal planar	120°
4	0	tetrahedral	109.5°
4	1	trigonal pyramidal	$\approx 107^\circ$
4	2	bent	$\approx 105^\circ$
5	0	trigonal bipyramidal	$90^\circ, 120^\circ$
6	0	octahedral	90°
6	1	square pyramidal	$\approx 90^\circ$
6	2	square planar	90°

“pyramidal”, or “bent”.

Exam wording often asks “planar”, “non-planar”, “bond angle”,

6.5.3 Planar molecule recognition

$$\text{linear, trigonal planar, square planar, and many aromatic rings are planar} \quad (6.6)$$

Tetrahedral and trigonal pyramidal centers are non-planar around that atom. A molecule is planar only when all atoms can lie in one plane.

6.5.4 Molecular polarity

$$\vec{\mu}_{\text{mol}} = \sum_i \vec{\mu}_i \quad (6.7)$$

Polar bonds can cancel in symmetric geometries. Lone pairs often prevent cancellation and make the molecule polar.

6.6 Exam recipe: molecular geometry, bond angle, and planarity

Use when: the problem asks for VSEPR shape, approximate bond angle, planar/non-planar, or central-atom geometry.

Given / Find: a molecular formula or Lewis structure; find geometry, requested angle, and planarity.

Required references: Equation 6.5, Table 6.2, and Equation 6.6.

Steps:

1. Draw or verify the Lewis structure, including central-atom lone pairs.
2. Count bonded atoms and lone pairs around each relevant center; compute SN by Equation 6.5.
3. Match SN and lone-pair count to Table 6.2.
4. Report the listed angle; state that lone pairs compress an ideal angle where applicable.
5. Test planarity from the local geometries: one tetrahedral or trigonal-pyramidal arrangement containing required atoms makes the molecule non-planar.

Checks: multiple bonds were counted as one domain; the stated geometry matches the lone-pair count; the planarity claim concerns all atoms requested.

Common traps: counting bond order as domains; reporting electron-domain geometry instead of molecular geometry; calling a molecule planar from one planar fragment.

6.7 Exam recipe: molecular polarity

Use when: the problem asks for polar/nonpolar, dipole moment, bond-dipole cancellation, or polarity ranking.

Given / Find: bond identities and molecular structure; find whether $\vec{\mu}_{\text{mol}}$ is zero or nonzero.

Required references: Equation 6.7, Equation 6.8, and Table 6.2.

Steps:

1. Identify polar bonds by electronegativity difference using Equation 6.8.
2. Obtain the three-dimensional molecular geometry from the Lewis structure and VSEPR table.
3. Draw bond-dipole directions toward more electronegative atoms; include asymmetry caused by substituent changes or lone pairs.
4. Add vectors by symmetry using Equation 6.7: complete cancellation means nonpolar; a residual vector means polar.

Checks: a molecule with no polar bonds is nonpolar; equal symmetrically arranged dipoles cancel; an unsymmetrical arrangement usually does not.

Common traps: deciding molecular polarity from a single bond; ignoring shape or lone pairs; assuming every symmetric-looking drawing is three-dimensionally symmetric.

6.8 Bonding formulas

6.8.1 Bond polarity from electronegativity

$$\Delta\chi = |\chi_A - \chi_B| \quad (6.8)$$

Larger electronegativity difference gives a more polar bond. Geometry decides whether the whole molecule is polar.

6.9 Exam recipe: ionic versus covalent bond character

Use when: the task asks which bond has greater ionic character, greater covalent character, or the smallest/largest polarity.

Given / Find: Given several bonded atom pairs and electronegativity values or periodic positions; rank bond character.

Required references: Equation (6.8); use supplied electronegativity values when provided.

Steps:

1. For each bonded pair, obtain χ_A and χ_B from the provided table or accepted periodic trend data.
2. Calculate $\Delta\chi = |\chi_A - \chi_B|$ for every pair using Equation (6.8).

- Rank increasing ionic character by increasing $\Delta\chi$; reverse this ranking for increasing covalent character.
- For a bond between identical atoms, set $\Delta\chi = 0$ and classify it as the most purely covalent option in that comparison.

Checks: The ranking is based on a bond, not whole-molecule polarity; larger $\Delta\chi$ cannot be labelled more covalent in the same set.

Common traps: using molecular shape to rank bond character; reversing the covalent ranking; overlooking that a homonuclear bond has zero electronegativity difference.

6.9.1 Molecular orbital bond order

$$\text{bond order} = \frac{N_{\text{bonding}} - N_{\text{antibonding}}}{2} \quad (6.9)$$

Higher bond order usually means shorter, stronger bonds. Zero bond order predicts no stable bond in simple MO treatment.

7 Intermolecular Forces, Solids, Liquids, and Colligative Properties

7.1 Ctrl+F keywords: vapor pressure, hydrogen bonding, boiling point, unit cell, BCC, FCC, density, freezing point depression, Raoult

Use for intermolecular force ranking, vapor pressure, crystal unit-cell density, cubic radius relations, and colligative-property calculations.

7.2 Symbols and notation

Table 7.1: Intermolecular forces, phase behavior, unit-cell, and colligative-property symbols.

Symbol	Meaning and usage
$P, \Delta H_{\text{vap}}, T, R, C$	Vapor pressure, enthalpy of vaporization, absolute temperature, gas constant, and integration constant.
P_1, P_2, T_1, T_2	Vapor pressures and temperatures at states 1 and 2.
X, Y	Electronegative atoms F, O, or N in the hydrogen-bond motif.
$\Delta H_{\text{fus}}, \Delta H_{\text{sub}}$	Enthalpy of fusion and sublimation.
Z, M, N_A, a, r, ρ	Units per cell, molar mass, Avogadro constant, edge length, atomic radius, and density.
$X_{\text{solvent}}, X_{\text{solute}}, P^\circ$	Mole fractions and pure-solvent vapor pressure.
m, i, K_f, K_b	Molality, van't Hoff factor, freezing-point constant, and boiling-point constant.
$\Delta T_f, \Delta T_b$	Freezing-point depression and boiling-point elevation.

7.3 Intermolecular forces and phase behavior

7.3.1 Hydrogen bonding recognition



Hydrogen bonding requires H bonded to F, O, or N. It raises boiling points and lowers vapor pressure relative to similar molecules.

7.3.2 Dispersion and dipole forces

$$\text{all molecules: dispersion,} \quad \text{polar molecules: dipole-dipole} \quad (7.2)$$

Larger molar mass and surface area increase dispersion forces. Stronger intermolecular forces usually mean higher boiling point and lower vapor pressure.

7.3.3 Like dissolves like

$$\text{polar dissolves polar,} \quad \text{nonpolar dissolves nonpolar} \quad (7.3)$$

Solubility is favored when solute-solvent interactions can replace solute-solute and solvent-solvent interactions.

7.4 Vapor pressure and phase changes

7.4.1 Clausius–Clapeyron equation

$$\ln P = -\frac{\Delta H_{\text{vap}}}{RT} + C \quad (7.4)$$

A plot of $\ln P$ versus $1/T$ has slope $-\Delta H_{\text{vap}}/R$. Vapor pressure increases with temperature.

7.4.2 Two-point vapor pressure equation

$$\ln\left(\frac{P_2}{P_1}\right) = -\frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad (7.5)$$

Use kelvin. This form calculates P_2 , T_2 , or ΔH_{vap} from two states.

7.4.3 Phase transition enthalpies

$$\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}} \quad (7.6)$$

Fusion, vaporization, and sublimation are endothermic in the forward direction.

7.5 Exam recipe: intermolecular force, vapor pressure, and boiling point

Use when: molecules are compared by intermolecular forces, volatility, vapor pressure, boiling point, or a vapor-pressure calculation is requested.

Given / Find: molecular structures or $P, T, \Delta H_{\text{vap}}$ data; find dominant force, ranking, unknown vapor pressure, temperature, or vaporization enthalpy.

Required references: Equation 7.1, Equation 7.2, Equation 7.4, and Equation 7.5.

Python: Brug `exam_toolbox.ipynb` med `clausius_clapeyron_pressure(...)` naar P_2 skal beregnes fra to temperaturer og ΔH_{vap} .

Steps:

1. For each molecule, mark dispersion, dipole–dipole behavior if polar, and hydrogen bonding only when the motif in Equation 7.1 is present.
2. For structurally comparable substances, rank interaction strength: hydrogen bonding and strong polarity first, then compare dispersion using size and surface area.
3. Convert the force ranking to properties: stronger forces imply lower vapor pressure, lower volatility, and higher boiling point.
4. If two vapor-pressure states are supplied, convert temperatures to kelvin and substitute in Equation 7.5; otherwise use Equation 7.4 for a line or slope question.
5. Solve for the one requested unknown and state its units.

Checks: increasing T must increase P ; a stronger-force compound should not be ranked as more volatile under comparable conditions; logarithm arguments are unitless ratios.

Common traps: claiming hydrogen bonding from an oxygen or nitrogen atom without an X–H donor; using Celsius in $1/T$; reversing vapor-pressure and boiling-point rankings.

7.6 Crystals and unit cells

7.6.1 Atoms per cubic unit cell

Table 7.2: Atoms or formula units and packing fractions for cubic unit cells.

Cell type	Z	Packing fraction	Corner atoms contribute 1/8, face atoms 1/2, and body-center atoms 1.
Simple cubic	1	0.52	
Body-centered cubic, BCC	2	0.68	
Face-centered cubic, FCC	4	0.74	

7.6.2 Unit-cell density

$$\rho = \frac{ZM}{N_A a^3} \quad (7.7)$$

Z is atoms or formula units per cell and a is cubic edge length. Convert a to centimeters if ρ is in g cm^{-3} .

7.6.3 Cubic radius relations

Table 7.3: Edge length and atomic-radius relations for cubic unit cells.

Cell type	Edge-radius relation	
Simple cubic	$a = 2r$	Use the diagonal that actually touches atoms: edge for simple cubic, body diagonal for BCC, face diagonal for FCC.
BCC	$\sqrt{3}a = 4r$	
FCC	$\sqrt{2}a = 4r$	

7.7 Exam recipe: unit-cell density, volume, and radius

Use when: the problem gives a simple cubic, BCC, or FCC solid together with density, atomic radius, molar mass, edge length, or unit-cell volume.

Given / Find: cell type and any two useful dimensional or mass quantities; find ρ , a^3 , a , or r .

Required references: Table 7.2, Table 7.3, and Equation 7.7.

Python: Brug `exam_toolbox.ipynb` med `unit_cell_volume(...)` naar densitet og Z skal omsættes til cellevolumen.

Steps:

1. Read Z for the identified cell type from Table 7.2.
2. Determine a : use $a^3 = V_{\text{cell}}$, or rearrange the matching relation in Table 7.3 when r is given.
3. Convert length to centimeters before cubing when density is required in g cm^{-3} .
4. Substitute Z , M , N_A , and a^3 in Equation 7.7; rearrange only for the requested unknown.
5. If radius is requested after solving for a , use the correct row of Table 7.3.

Checks: $1 \text{ pm} = 10^{-10} \text{ cm}$; $1 \text{ nm} = 10^{-7} \text{ cm}$; a density answer has mass per volume units.

Common traps: using the FCC relation for BCC; omitting Z ; cubing a length before converting its units.

7.8 Colligative properties

7.8.1 Raoult's law and vapor-pressure lowering

$$P_{\text{solvent}} = X_{\text{solvent}} P_{\text{solvent}}^{\circ} \quad (7.8)$$

$$\Delta P = X_{\text{solute}} P_{\text{solvent}}^{\circ} \quad (7.9)$$

A nonvolatile solute lowers solvent vapor pressure in proportion to solute mole fraction.

7.8.2 Molality and freezing point depression

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}}(\text{kg})} \quad (7.10)$$

$$\Delta T_f = i K_f m \quad (7.11)$$

$$\Delta T_b = i K_b m \quad (7.12)$$

$$T_{f,\text{solution}} = T_f^{\circ} - \Delta T_f, \quad T_{b,\text{solution}} = T_b^{\circ} + \Delta T_b \quad (7.13)$$

$$n_{\text{solute}} = m m_{\text{solvent}}(\text{kg}), \quad M_{\text{solute}} = \frac{m_{\text{solute}}}{n_{\text{solute}}} \quad (7.14)$$

i is the van't Hoff factor, approximately the number of dissolved ions for complete dissociation.

7.9 Exam recipe: freezing or boiling point change

Use when: the problem gives solute mass, solvent mass, K_f , K_b , dissociation information, or asks for a new freezing or boiling temperature.

Given / Find: solute identity and amount, solvent mass and reference phase-change temperature; find m , ΔT_f , ΔT_b , or the solution phase-change temperature.

Required references: Equations 7.10, 7.11, 7.12, and 7.13.

Python: Brug `exam_toolbox.ipynb` med `freezing_point_depression(...)` til frysepunktssænkning med solut, solventmasse og van't Hoff-faktor.

Steps:

1. Convert the solute amount to moles and convert solvent mass alone to kilograms.
2. Calculate molality with Equation 7.10.
3. Choose $i = 1$ for a nonelectrolyte or use the stated/ideal dissociation count for an electrolyte.
4. Use Equation 7.11 or Equation 7.12 as requested.
5. Apply the calculated change to the pure-solvent value with Equation 7.13.

Checks: a nonvolatile solute lowers freezing point and raises boiling point; m uses kilograms of solvent; ΔT is reported as a positive magnitude before applying its direction.

Common traps: using total solution mass as solvent mass; forgetting i for a salt such as NaBr; subtracting a boiling-point elevation.

7.10 Exam recipe: molar mass from freezing or boiling point change

Use when: an unknown solute mass and an observed freezing-point depression or boiling-point elevation are given and its molar mass is requested.

Given / Find: Given m_{solute} , solvent mass, K_f or K_b , measured phase-change temperature, and dissociation assumption; find M_{solute} .

Required references: Equations 7.11, 7.12, 7.13, and 7.14.

Python: Brug `exam_toolbox.ipynb` med `freezing_point_depression(...)` som kontrol for m , ΔT_f og frysepunkt; omarranger stadig manuelt naar den ukendte er molarmassen.

Steps:

1. Calculate the positive temperature change as pure-solvent freezing point minus solution freezing point, or solution boiling point minus pure-solvent boiling point, using Equation 7.13.
2. Select $i = 1$ for an indicated nonelectrolyte or use the stated dissociation factor, then rearrange the matching colligative equation to $m = \Delta T / (iK)$.
3. Convert solvent mass to kilograms and calculate solute moles with the first relation in Equation 7.14.
4. Divide the stated solute mass by the calculated moles using the second relation in Equation 7.14.

Checks: ΔT , molality, moles, and molar mass must all be positive; a smaller observed change for the same solute mass implies a larger molar mass.

Common traps: using solution mass as solvent mass; choosing the wrong sign for ΔT ; assuming electrolyte dissociation for an organic nonelectrolyte.

8 Kinetics and Chemical Equilibrium

8.1 Ctrl+F keywords: rate law, reaction order, first order, second order, half-life, Arrhenius, catalyst, equilibrium, Le Chatelier, K_p

Use for concentration-time data, identifying order, half-life, activation energy, catalyst statements, equilibrium expressions, Q versus K , and gas equilibria.

8.2 Symbols and notation

Table 8.1: Kinetics and equilibrium symbols

Symbol	Meaning and usage
$r, k, t, [A]_t$	Reaction rate, rate constant, time, and concentration of A at time t .
$\frac{d[A]}{dt}$	Change in concentration of species A per unit time.
a, b, c, d	Stoichiometric coefficients in the example reaction.
m, n	Experimentally determined orders in the rate law; not amounts of substance here.
$t_{1/2}, E_a, A, R, T$	Half-life, activation energy, frequency factor, gas constant, and temperature.
cat, uncat	Subscripts for catalyzed and uncatalyzed pathways.
Q, K, a_A	Reaction quotient, equilibrium constant, and thermodynamic activity.
$\alpha, \beta, \gamma, \delta$	Stoichiometric coefficients in the general equilibrium reaction.
$K_p, K_c, p_A, \Delta n_{\text{gas}}$	Pressure/concentration equilibrium constants, partial pressure, and gas coefficient change.

8.3 Kinetics formulas

8.3.1 Reaction rate from stoichiometry



$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt} \quad (8.2)$$

Divide by coefficients so every species reports the same reaction rate.

8.3.2 Rate law and overall order

$$r = k[A]^m[B]^n, \quad \text{overall order} = m + n \quad (8.3)$$

Orders are found experimentally unless the problem states an elementary step.

8.3.3 Initial-rate comparison

$$\frac{r_2}{r_1} = \left(\frac{[A]_2}{[A]_1} \right)^m \left(\frac{[B]_2}{[B]_1} \right)^n \quad (8.4)$$

Compare experiments in which all but one reactant concentration are unchanged to isolate one order at a time.

8.3.4 Integrated rate laws and linearity test

Table 8.2: Transformed concentration-time tests for reaction order

Order	Transform to plot against t	Linear relation	Slope gives k	The transformed data set that forms a straight line identifies the order; its fitted slope gives k .
0	$[A]_t$	$[A]_t = -kt + [A]_0$	$k = -\text{slope}$	
1	$\ln[A]_t$	$\ln[A]_t = -kt + \ln[A]_0$	$k = -\text{slope}$	
2	$1/[A]_t$	$1/[A]_t = kt + 1/[A]_0$	$k = \text{slope}$	

8.3.5 Half-life

$$t_{1/2}^{(0)} = \frac{[A]_0}{2k}, \quad t_{1/2}^{(1)} = \frac{\ln 2}{k}, \quad t_{1/2}^{(2)} = \frac{1}{k[A]_0} \quad (8.5)$$

Only first-order half-life is independent of initial concentration.

8.3.6 Arrhenius equation

$$k = Ae^{-E_a/RT}, \quad \ln k = -\frac{E_a}{R} \frac{1}{T} + \ln A \quad (8.6)$$

A plot of $\ln k$ versus $1/T$ has slope $-E_a/R$.

8.3.7 Two-temperature Arrhenius relation

$$\ln\left(\frac{k_2}{k_1}\right) = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad (8.7)$$

Use kelvin. If $T_2 > T_1$, then normally $k_2 > k_1$.

8.3.8 Catalyst effect

$$E_{a,cat} < E_{a,uncat}, \quad K_{eq,cat} = K_{eq,uncat}, \quad \Delta G_{cat}^\circ = \Delta G_{uncat}^\circ \quad (8.8)$$

A catalyst speeds forward and reverse reactions by lowering activation barriers; it does not move equilibrium.

8.4 Exam recipe: reaction order from concentration-time data

Use when: a table or graph gives $[A]$ versus time and the task asks for order, k , or a concentration prediction.

Given / Find: given t and $[A]_t$; find zero-, first-, or second-order behavior and the corresponding k .

Required references: Table 8.2 and equation (8.5) when a half-life is requested.

Python: Brug `exam_toolbox.ipynb` med `kinetic_linear_fit(...)` for hver af orden 0, 1 og 2, saa R^2 , hældning og skæring sammenlignes ensartet.

Steps:

1. Construct all three transformed data sets by calculating $[A]_t$, $\ln[A]_t$, and $1/[A]_t$ for each listed time.
2. Determine the order by plotting each transformed data set against t , or by comparing successive slopes, and choosing the one with approximately constant slope (best linearity).
3. Obtain k from the linear fit slope by using $k = -\text{slope}$ for the $[A]_t$ or $\ln[A]_t$ line and $k = \text{slope}$ for the $1/[A]_t$ line.
4. Predict a missing concentration or time by inserting the fitted k , the initial value, and the requested variable into the selected linear relation in Table 8.2.

Checks: verify that $k > 0$, that only one transformation is clearly linear within data precision, and that first-order data show approximately equal fractional loss over equal time intervals.

Common traps: selecting an order from the raw decreasing curve without testing transformations; using the negative slope itself as k for order zero or one; taking $\ln$ of a value with inconsistent concentration units.

8.5 Exam recipe: reaction order from initial-rate experiments

Use when: a data table gives initial concentrations and initial rates for multiple experimental runs.

Given / Find: Given initial-rate runs; find the order in each reactant, overall order, or the rate constant.

Required references: Equations (8.3) and (8.4).

Steps:

1. Choose two runs in which every concentration except one reactant is identical; this makes all unchanged concentration ratios equal to one in Equation (8.4).
2. Calculate the remaining concentration ratio and rate ratio, then solve $\text{rate ratio} = (\text{concentration ratio})^{\text{order}}$; for ratios not obvious by inspection, use $\text{order} = \log(r_2/r_1)/\log(c_2/c_1)$.
3. Repeat step 1 for each reactant whose order is required and add the exponents for overall order using Equation (8.3).
4. If k is requested, substitute one complete experimental run and the determined exponents into Equation (8.3), then isolate k .

Checks: Substitution of a second run should reproduce its rate; unchanged concentrations must not affect the isolated exponent.

Common traps: comparing runs where two concentrations change before both orders are known; treating stoichiometric coefficients as measured orders; reversing numerator and denominator in only one ratio.

8.6 Exam recipe: Arrhenius relation and catalyst

Use when: the task gives rate constants at temperatures, asks for E_a , or asks what a catalyst changes.

Given / Find: given k and T values and/or catalyst information; find E_a , a new k , or the correct catalyst conclusion.

Required references: equations (8.6), (8.7), and (8.8).

Python: Brug `exam_toolbox.ipynb` med `arrhenius_ratio(...)` naar opgaven spoerger efter k_2/k_1 .

Steps:

1. Prepare temperatures by converting each temperature to kelvin before calculating $1/T$ or using a temperature ratio relation.
2. Find E_a from two measurements by substituting k_1, k_2, T_1, T_2 into equation (8.7) and isolating E_a , or from several measurements by fitting $\ln k$ versus $1/T$ and using $E_a = -R(\text{slope})$.
3. Find an unknown rate constant by rearranging equation (8.7) after E_a is known.
4. Evaluate a catalyst statement by applying equation (8.8): lower activation energy and faster approach to equilibrium, but unchanged K_{eq} and ΔG° .

Checks: for a positive E_a , raising T must raise k ; activation energy should have energy-per-mole units.

Common traps: using degrees Celsius in reciprocal temperature; losing the minus sign in the Arrhenius slope; claiming that a catalyst changes equilibrium composition.

8.7 Equilibrium formulas

8.7.1 Reaction quotient and equilibrium constant



$$Q = \frac{a_C^\gamma a_D^\delta}{a_A^\alpha a_B^\beta}, \quad Q_{\text{eq}} = K \quad (8.10)$$

Pure solids, pure liquids, and the solvent are omitted from the expression.

8.7.2 Direction from Q and K

$$Q < K : \text{forward shift}, \quad Q = K : \text{equilibrium}, \quad Q > K : \text{reverse shift} \quad (8.11)$$

The system shifts so that Q approaches K .

8.7.3 Gas equilibrium K_p

$$K_p = \frac{(p_C)^\gamma (p_D)^\delta}{(p_A)^\alpha (p_B)^\beta} \quad (8.12)$$

Use partial pressures with stoichiometric powers and omit pure condensed phases.

8.7.4 Relation between K_p and K_c

$$K_p = K_c (RT)^{\Delta n_{\text{gas}}}, \quad \Delta n_{\text{gas}} = \sum \nu_{\text{gas, products}} - \sum \nu_{\text{gas, reactants}} \quad (8.13)$$

8.7.5 Le Chatelier shift rules

Table 8.3: Quick equilibrium shift decisions

Disturbance	Shift decision
Add reactant or remove product	Toward products
Remove reactant or add product	Toward reactants
Decrease volume for gas system	Toward fewer moles of gas
Raise T for an endothermic forward reaction	Toward products
Raise T for an exothermic forward reaction	Toward reactants
Add catalyst	No equilibrium shift

8.8 Exam recipe: equilibrium expression and shift

Use when: the task asks for K , Q , K_p , reaction direction, or a Le Chatelier/catalyst statement.

Given / Find: given a balanced equilibrium and activities, concentrations, partial pressures, or a disturbance; find the equilibrium expression, Q -based direction, or qualitative shift.

Required references: equations (8.10), (8.11), (8.12), and (8.13); Table 8.3.

Steps:

1. Write the expression by placing product activities over reactant activities and using balanced stoichiometric coefficients as exponents, while omitting pure solids and pure liquids.
2. Choose the quantity type by using partial pressures in equation (8.12) for K_p , concentrations for K_c , or converting with equation (8.13) when required.
3. Determine the spontaneous direction from measured values by calculating Q with the same expression as K and comparing it with K using equation (8.11).
4. Determine the response to an imposed change by locating the disturbance in Table 8.3; for temperature changes, treat heat as a reactant for an endothermic forward reaction and as a product for an exothermic one; for a catalyst, state faster equilibration rather than a shifted position.

Checks: exponents must match the balanced equation; $Q = K$ means no net shift; a volume change only favors a side if the gaseous mole totals differ.

Common traps: including a pure solid or liquid in K ; reversing the $Q < K$ direction; stating that a catalyst changes K .

8.8.1 ICE-table extent relation

$$n_i = n_{i,0} + \nu_i \xi, \quad [i] = \frac{n_i}{V} \quad (8.14)$$

Use negative signed coefficients ν_i for reactants and positive coefficients for products in the forward direction.

8.9 Exam recipe: equilibrium composition from K and initial amounts

Use when: initial amounts or concentrations and an equilibrium constant are given, and the equilibrium amount or concentration is requested.

Given / Find: Given a balanced equilibrium, initial composition, volume, and K or thermodynamic data from which K can be obtained; find the equilibrium composition.

Required references: Equations (8.10), (8.14), and (4.10) when K must first be calculated.

Python: Brug `exam_toolbox.ipynb` med `equilibrium_constant(...)` naar K foerst skal findes fra ΔG° ; selve ICE-polynomiet skal opstilles fra reaktionen foer numerisk loesning.

Steps:

1. Balance the equilibrium and write the K expression from Equation (8.10); omit pure solids and liquids.
2. Obtain K : use the supplied value directly or calculate it from ΔG° with Equation (4.10).
3. Define one forward extent ξ and write every equilibrium amount as $n_{i,0} + \nu_i \xi$ using Equation (8.14); divide by common volume when a concentration expression is needed.
4. Substitute those expressions into K , solve for ξ , and reject any mathematical root that makes any physical amount negative.
5. Insert the accepted ξ into each requested species expression and report amount or concentration with units.

Checks: Equilibrium amounts are nonnegative and give the stated K when substituted back; the sign of ξ agrees with the initial Q -versus- K direction.

Common traps: using initial concentrations directly as equilibrium values; omitting stoichiometric multipliers in the ICE row; retaining a nonphysical algebraic root.

9 Organic Chemistry, Functional Groups, Chirality, and Polymers

9.1 Ctrl+F keywords: functional group, alcohol, ether, ester, amide, carboxylic acid, aromatic, chirality, isomer, polymerization

Use for identifying organic functional groups, aromatic rings, isomers, chiral centers, amino acids, peptides, and polymer repeat units.

9.2 Symbols and notation

Table 9.1: Symbols and notation for organic chemistry.

Symbol	Meaning and usage
n	Number of carbon atoms in general hydrocarbon formulas; polymerization degree when used with polymer masses.
C_nH_{2n+2}	General formula for an acyclic alkane.
C_nH_{2n} , C_nH_{2n-2}	General formulas for acyclic alkenes and alkynes.
DBE	Degree of unsaturation: total rings and π -bonds.
C , H , N , X	Atom counts in the DBE formula; X counts halogens.
R , R'	Generic organic groups; primes distinguish groups that need not be identical.
C^*	A tetrahedral stereogenic carbon attached to four different groups.
M_{polymer} , M_{repeat}	Polymer molar mass and repeat-unit molar mass.

9.3 Hydrocarbons and unsaturation

9.3.1 Alkane, alkene, alkyne formulas

$$\text{alkane: } C_nH_{2n+2}, \quad \text{alkene: } C_nH_{2n}, \quad \text{alkyne: } C_nH_{2n-2} \quad (9.1)$$

Alkenes contain $C=C$ and alkynes contain $C\equiv C$. Rings reduce hydrogen count like one double bond.

9.3.2 Degree of unsaturation, double bond equivalent

$$\text{DBE} = \frac{2C + 2 + N - H - X}{2} \quad (9.2)$$

Oxygen does not affect DBE. Each ring or π -bond counts as one DBE; benzene has four DBE.

9.3.3 Aromatic ring recognition

$$C_6H_6 \text{ ring motif}, \quad \text{phenyl group: } C_6H_5- \quad (9.3)$$

A benzene ring is planar and aromatic. Aromatic rings often occur with alcohol, carboxylic acid, amide, or ester groups.

9.4 Functional group lookup

9.4.1 Functional group recognition table

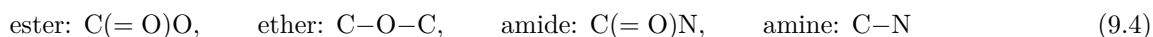
Table 9.2: Functional groups recognized from structural motifs.

Group	Structural motif
alcohol	$R-OH$
ether	$R-O-R'$
aldehyde	$R-CHO$
ketone	$R-C(=O)-R'$
carboxylic acid	$R-COOH$
ester	$R-C(=O)-O-R'$
amine	$R-NH_2$, R_2NH , R_3N
amide	$R-C(=O)-NH_2$, $R-C(=O)-NR_2$
nitrile	$R-C\equiv N$

directly to the carbonyl carbon.

Carbonyl-containing groups are distinguished by what is attached

9.4.2 Ester, ether, amide, amine distinction



Look for the carbonyl first. A nitrogen next to a carbonyl is an amide, not an amine.

9.5 Exam recipe: functional groups in a molecule

Use when: an organic structure is shown and the functional groups present must be identified.

Given / Find: Given a line, condensed, or displayed structure; find every functional group named in the question or answer options.

Required references: Table 9.2 and Equation (9.4); use Equation (9.3) for aromatic rings.

Steps:

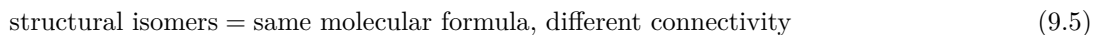
1. Mark every heteroatom O, N, halogen, triple bond, and carbonyl C=O.
2. For each carbonyl carbon, inspect its directly attached atom and classify acid, ester, amide, aldehyde, or ketone.
3. Classify remaining O and N motifs as alcohol, ether, or amine only after carbonyl groups are assigned.
4. Identify aromatic rings and nitriles separately.
5. Match the complete list against the requested response or eliminate options containing a missing motif.

Checks: Every reported group must correspond to an explicit motif; an acid counts as carboxylic acid, not separate alcohol plus ketone.

Common traps: Calling an ester an ether; calling an amide an amine; overlooking a carbonyl or an aromatic ring.

9.6 Isomerism, chirality, and biomolecules

9.6.1 Structural isomers



Structural isomers can have different names, boiling points, and functional groups.

9.7 Exam recipe: molecular formula consistency and constitutional isomer enumeration

Use when: a displayed structure must be tested against a molecular formula, or the number of constitutional isomers with a stated functional group is requested.

Given / Find: Given a molecular formula, candidate structures, or a functional-group restriction such as alcohol; find inconsistent structures or the number of distinct connectivities.

Required references: Equations (9.2), (9.5), and Table 9.2.

Steps:

1. Calculate DBE from Equation (9.2); use it to determine how many total rings and multiple bonds a valid candidate can contain.
2. To test a shown option, count each element in the drawing and compare with the stated formula, then reject it if the atom count or DBE does not match.
3. To enumerate isomers, first list distinct carbon skeletons compatible with the formula and DBE, without placing the required functional group twice on symmetry-equivalent positions.
4. Place the required functional group on each nonequivalent permitted position of each skeleton; for an alcohol, attach -OH to carbon and exclude ethers unless the question includes them.
5. Count only structures with different connectivity as required by Equation (9.5); do not count mirror images or rotations as additional constitutional isomers.

Checks: Every counted structure has the stated formula, DBE, and required functional group; for saturated acyclic $\text{C}_4\text{H}_{10}\text{O}$ alcohols, DBE must be zero.

Common traps: counting an ether among alcohols; counting the same structure redrawn from another end; forgetting a branched carbon skeleton.

9.7.1 Chiral center

$$C^* = \text{tetrahedral carbon attached to four different groups} \quad (9.6)$$

A molecule with one chiral center is chiral. Symmetry can make molecules with multiple stereocenters achiral.

9.7.2 Amino acids and peptide bond

$$\text{amino acid motif: } H_2N-CHR-COOH \quad (9.7)$$

$$\text{peptide bond: } C(=O)-NH \quad (9.8)$$

Amino acids contain amine and carboxylic acid groups. Peptides are connected by amide bonds.

9.8 Exam recipe: chirality and peptide or biomolecule recognition

Use when: a structure must be classified as chiral, amino-acid-like, or peptide-containing.

Given / Find: Given a molecular structure; find stereogenic carbons and identify amino acid or peptide motifs.

Required references: Equations (9.6), (9.7), and (9.8); Table 9.2.

Steps:

1. Locate tetrahedral carbon candidates and list the four attached groups for each candidate.
2. Mark C^* only when all four groups differ; if multiple centers occur, check for internal symmetry before concluding that the molecule is chiral.
3. Search for the amino acid pattern: amine and carboxylic acid attached through the same alpha-carbon framework.
4. Search for $C(=O)-NH$ links between residues; classify each such link as a peptide (amide) bond.
5. State only the requested property: number of chiral centers, chirality, amino acid motif, or peptide bond count.

Checks: A carbon with two identical substituents is not stereogenic; a peptide bond must include a carbonyl directly bonded to nitrogen.

Common traps: Counting a planar carbonyl carbon as chiral; calling any $C-N$ bond a peptide bond; treating an amide nitrogen as an amine group.

9.9 Polymers

9.9.1 Degree of polymerization

$$n = \frac{M_{\text{polymer}}}{M_{\text{repeat}}} \quad (9.9)$$

The degree of polymerization is the number of repeat units in the chain.

9.10 Exam recipe: polymer repeat unit and degree of polymerization

Use when: a polymer structure, monomer, repeat-unit mass, or polymer molar mass is given.

Given / Find: Given the repeat-unit structure or its molar mass and M_{polymer} ; find the repeat unit, M_{repeat} , or degree n .

Required references: Equation (9.9) and the atomic molar masses supplied in the problem or data sheet.

Python: Brug `exam_toolbox.ipynb` med `molar_mass(...)` til repeat-unit-massen, naar repeat unit kan skrives som en sumformel.

Steps:

1. Identify the smallest structural segment that repeats along the chain and write its atom count.
2. Compute M_{repeat} from that segment unless it is supplied.
3. Insert consistent molar-mass units into Equation (9.9).
4. Calculate n and round only as justified by the idealized chain or stated precision.

Checks: n is dimensionless and is normally close to a positive integer in an idealized exam problem; multiplying nM_{repeat} should recover M_{polymer} .

Common traps: Using monomer mass when condensation removes a small molecule; counting end groups as part of every repeat unit; reporting n with mass units.

10 Acids, Bases, Buffers, Titration, and Solubility Products

10.1 Ctrl+F keywords: pH, pOH, weak acid, K_a , pK_a , buffer, Henderson, phosphate, titration, K_{sp} , common ion, oploeselighed

Use for acid strength, pH calculations, buffers, phosphate systems, titration equivalence, solubility, precipitation, and common-ion tasks.

10.2 Symbols and notation

Table 10.1: Acid-base and solubility symbols

Symbol	Meaning and usage
K_w , K_a , K_b	Water autoionization, acid dissociation, and base dissociation constants.
pH, pOH, pK_a , pK_b	Negative base-10 logarithms of hydronium, hydroxide, K_a , and K_b .
HA, A^- , B, HB^+	Weak acid/conjugate base and weak base/conjugate acid notation.
c_0 , x	Initial analytical concentration and amount dissociated per volume in an equilibrium setup.
K_{sp} , Q_{sp} , s	Solubility-product constant, current ion product, and molar solubility.
M, X	Generic cation/metal and anion in a sparingly soluble salt.
a , b , m , x	Stoichiometric subscripts or ionic charges in M_aX_b ; identified by context.

10.3 pH and acid-base constants

10.3.1 Water, pH, and pOH

$$K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}, \quad pH = -\log[H_3O^+], \quad pOH = -\log[OH^-], \quad pH + pOH = 14 \quad (10.1)$$

At 298 K, neutral water has pH 7. A decrease of one pH unit means ten times larger $[H_3O^+]$.

10.3.2 Strong acid and strong base pH

$$[H_3O^+] \approx \nu_{H^+} c_{acid}, \quad [OH^-] \approx \nu_{OH^-} c_{base} \quad (10.2)$$

Strong acids and bases are treated as fully dissociated. Include stoichiometric coefficients; for example, $Ba(OH)_2$ gives $2OH^-$.

10.3.3 Weak acid and weak base constants

$$K_a = \frac{[A^-][H_3O^+]}{[HA]}, \quad K_b = \frac{[HB^+][OH^-]}{[B]} \quad (10.3)$$

$$K = 10^{-pK}, \quad pK = -\log K \quad (10.4)$$

Larger K_a or smaller pK_a means stronger acid. Larger K_b or smaller pK_b means stronger base.

Table 10.2: Common acid and base identities for qualitative classification when no K_a data are supplied

Category	Course-level examples	Classification action
Strong acids	HCl, HNO_3 , first proton of H_2SO_4	Rank above listed weak acids in water.
Weak acids	HF, CH_3COOH , HNO_2 , H_3PO_4	Acidic, but weaker than the listed strong acids.
Bases / non-acids	NaOH, NH_3 , metal hydrides such as NaH	Reject from an option requiring acids only.
Generally non-acid organic compounds	hydrocarbons and ordinary alcohols without an acidic group	Do not classify as common aqueous acids unless data state otherwise.

10.3.4 Conjugate acid-base relation

$$K_a K_b = K_w, \quad pK_a + pK_b = 14 \quad (10.5)$$

Use this for conjugate acid/base pairs at 298 K.

10.3.5 Weak acid and weak base approximation with exact fallback

$$[\text{H}_3\text{O}^+] \approx \sqrt{K_a c_0}, \quad [\text{OH}^-] \approx \sqrt{K_b c_0} \quad (10.6)$$

$$K = \frac{x^2}{c_0 - x} \implies x = \frac{-K + \sqrt{K^2 + 4Kc_0}}{2} \quad (10.7)$$

Use the approximation only when the computed $x/c_0 \leq 0.05$; otherwise use the exact positive root.

10.4 Exam recipe: identify acids and rank acid strength

Use when: answer options list chemical formulas and ask which is an acid, which set contains only acids or weak acids, or which acid is strongest without supplying numerical constants.

Given / Find: Given formulas and possibly K_a or pK_a data; find acid/non-acid classification or relative acid strength.

Required references: Table 10.2 and Equation (10.4) when numerical K_a or pK_a data are given.

Steps:

1. Match each listed species to Table 10.2; reject an “all acids” or “all weak acids” option as soon as it contains a listed base or non-acid.
2. For qualitative ranking without data, rank a listed strong acid above a listed weak acid; compare weak acids further only when course data or a supplied table supports it.
3. When K_a values are supplied, rank acids by decreasing K_a ; when pK_a values are supplied, rank them by increasing pK_a , using Equation (10.4).
4. State the classification or selected option and identify the specific species or constant comparison that decides it.

Checks: Strong acid means essentially complete dissociation for ordinary introductory calculations; smaller pK_a must correspond to larger K_a .

Common traps: assuming every compound containing hydrogen is an acid; classifying NH_3 as an acid because it contains hydrogen; ranking weak acids without supplied or established comparative data.

10.5 Exam recipe: weak acid or weak base pH with approximation and exact fallback

Use when: a weak acid/base concentration and K_a , K_b , pK_a , or pK_b are given without a pre-existing buffer pair.

Given / Find: given c_0 and an acid/base constant; find $[\text{H}_3\text{O}^+]$ or $[\text{OH}^-]$, then pH.

Required references: equations (10.1), (10.3), (10.4), (10.5), (10.6), and (10.7).

Python: Brug `exam_toolbox.ipynb` med `weak_solution_ph(...)` med "acid" eller "base" naar 5%-testen fejler eller pH skal kontrolleres hurtigt.

Steps:

1. Select the equilibrium constant by using K_a for HA or K_b for B, converting a supplied pK with Equation (10.4), and using Equation (10.5) only when the conjugate constant is supplied.
2. Estimate dissociation by calculating $x_{\text{approx}} = \sqrt{Kc_0}$ from equation (10.6), interpreting x as $[\text{H}_3\text{O}^+]$ for an acid and $[\text{OH}^-]$ for a base.
3. Decide whether the estimate is acceptable by calculating x_{approx}/c_0 ; retain it when the ratio is at most 0.05, otherwise calculate x from equation (10.7).
4. Report pH by evaluating $-\log[\text{H}_3\text{O}^+]$ for an acid or by evaluating pOH from $[\text{OH}^-]$ and subtracting it from 14 for a base using equation (10.1).

Checks: $0 < x < c_0$; a weak acid solution should be acidic and a weak base solution basic under ordinary concentrations; the approximation and exact result should be close when the 5% test passes.

Common traps: using the square-root approximation after it fails the 5% test; reporting pOH as pH for a weak base; mixing K_a with a base calculation without converting through K_w .

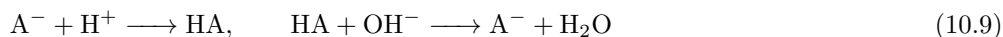
10.6 Buffers and titration

10.6.1 Henderson-Hasselbalch buffer equation

$$\text{pH} = \text{p}K_a + \log \left(\frac{[\text{A}^-]}{[\text{HA}]} \right) = \text{p}K_a + \log \left(\frac{n_{\text{A}^-}}{n_{\text{HA}}} \right) \quad (10.8)$$

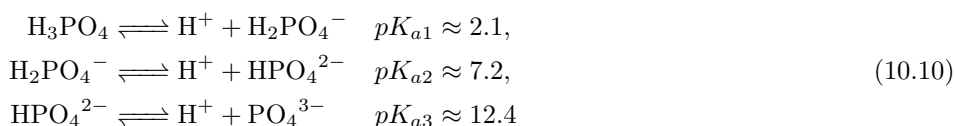
The mole ratio form applies when acid and base occupy the same final solution volume.

10.6.2 Buffer after adding strong acid or base



Complete the stoichiometric neutralization before applying the buffer equation.

10.6.3 Phosphoric acid phosphate system



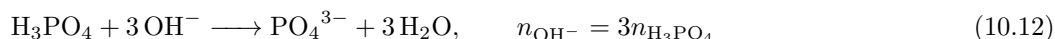
Choose the conjugate pair whose $\text{p}K_a$ is closest to the target or observed pH.

10.6.4 Titration equivalence point

$$n_{\text{acid equivalents}} = n_{\text{base equivalents}} \quad (10.11)$$

At equivalence, the pH depends on the acid-base properties of the salt remaining after neutralization.

10.6.5 Complete neutralization of a polyprotic acid



To form fully deprotonated phosphate, supply one mole of hydroxide for each acidic proton removed.

10.7 Exam recipe: acid-base stoichiometry and titration amount

Use when: the task asks for the amount or volume of strong acid/base needed to neutralize an acid/base or to form a specified phosphate species.

Given / Find: Given concentrations, volumes, and target neutralization product; find moles or volume of titrant.

Required references: Equations (10.11) and (10.12); use $n = cV$ from Equation (3.1).

Steps:

1. Identify how many protons or hydroxides must react per formula unit to produce the named target species; for sodium phosphate from H_3PO_4 , use the 3:1 hydroxide-to-acid ratio in Equation (10.12).
2. Calculate initial acid or base moles from its concentration and volume.
3. Multiply by the required stoichiometric equivalents to find titrant moles, using Equation (10.11).
4. Divide required titrant moles by titrant concentration to obtain the volume.

Checks: Conversion to fully deprotonated PO_4^{3-} requires three hydroxide equivalents per H_3PO_4 ; use liters in $n = cV$.

Common traps: stopping at the first equivalence point when the target salt requires full deprotonation; using milliliters without converting to liters.

10.8 Exam recipe: buffer pH

Use when: both a weak acid and its conjugate base are present, including a phosphate buffer or a buffer after addition of strong acid/base.

Given / Find: given pair amounts, $\text{p}K_a$, and optionally added acid/base; find buffer pH or the required ratio.

Required references: equations (10.8), (10.9), and (10.10) for phosphate buffers.

Steps:

1. Identify the conjugate pair by matching species that differ by one proton, and for phosphate choose the pair in equation (10.10) with pK_a nearest the relevant pH.
2. Convert the supplied amounts to moles by multiplying each concentration by its solution volume before any mixing or reaction.
3. Account for added strong acid or base by subtracting and adding moles according to equation (10.9) until the limiting strong reagent is consumed.
4. Calculate pH by inserting the remaining mole ratio n_{A^-}/n_{HA} into equation (10.8), or rearrange that equation to find a requested ratio.

Checks: equal conjugate-pair amounts give $\text{pH} = pK_a$; a valid buffer retains positive amounts of both pair members; adding acid must lower the pH.

Common traps: applying Henderson-Hasselbalch before neutralizing added strong reagent; selecting the wrong phosphate pK_a ; using concentrations from unequal pre-mixing volumes instead of post-reaction mole ratios.

10.9 Solubility products

10.9.1 Solubility product expression



$$K_{sp} = [M^{m+}]^a [X^{x-}]^b \quad (10.14)$$

The pure solid is omitted. Powers come from dissolution stoichiometry.

10.9.2 Molar solubility patterns

Table 10.3: Common molar-solubility substitutions

Solid	Ion concentrations from molar solubility s	K_{sp} relation
MX	$[M^+] = s, [X^-] = s$	$K_{sp} = s^2$
MX ₂	$[M^{2+}] = s, [X^-] = 2s$	$K_{sp} = 4s^3$
M(OH) ₂	$[M^{2+}] = s, [OH^-] = 2s$	$K_{sp} = 4s^3$

10.9.3 Ion product and precipitation

$$Q_{sp} = [M^{m+}]^a [X^{x-}]^b \quad (10.15)$$

Table 10.4: Precipitation decision from Q_{sp} and K_{sp}

Comparison	State or direction
$Q_{sp} < K_{sp}$	Unsaturated; additional solid can dissolve
$Q_{sp} = K_{sp}$	Saturated solution at equilibrium
$Q_{sp} > K_{sp}$	Supersaturated; precipitation is favored

10.9.4 Common-ion solubility

$$K_{sp} = [M^{m+}][X^{x-}], \quad [X^{x-}]_{\text{initial}} \gg s \implies s \approx \frac{K_{sp}}{[X^{x-}]_{\text{initial}}} \quad (10.16)$$

A common ion lowers solubility by shifting dissolution toward the solid.

10.10 Exam recipe: solubility product, common ion, and hydroxide pH

Use when: the task asks for molar solubility, precipitation, saturated-solution pH, or common-ion effects.

Given / Find: given a salt formula, K_{sp} , and optionally initial ion concentrations; find s , Q_{sp} , precipitation direction, or pH for a hydroxide.

Required references: equations (10.13), (10.14), (10.15), (10.16), and (10.1); Tables 10.3 and 10.4.

Python: Brug `exam_toolbox.ipynb` med `solubility_from_ksp(...)` til molar solubility med stoikiometri og common ion; brug `solubility_complete_check(...)` naar en tilsat saltmasse skal vurderes som helt opløst eller i overskud.

Steps:

1. Write the dissolution equation by splitting one formula unit into ions and preserving each stoichiometric subscript as an ion coefficient.
2. Build the K_{sp} or Q_{sp} expression by raising each dissolved ion concentration to its coefficient as in equation (10.14).
3. Calculate molar solubility by replacing each newly formed ion concentration with its coefficient times s , using Table 10.3 directly for the listed patterns and adding any given initial common-ion concentration before solving.
4. Decide whether precipitation occurs by calculating Q_{sp} from current concentrations and comparing it with K_{sp} using Table 10.4.
5. Find hydroxide-solution pH by converting the solved s into $[\text{OH}^-]$ through the dissolution coefficient, then calculating pOH and pH with equation (10.1).

Checks: a common ion must reduce s relative to pure water; for $\text{M}(\text{OH})_2$, $[\text{OH}^-] = 2s$; precipitation begins at $Q_{sp} = K_{sp}$.

Common traps: including the solid in K_{sp} ; missing powers generated by stoichiometric coefficients; setting $[\text{OH}^-] = s$ for a dihydroxide; applying the common-ion approximation when the initial ion concentration is not much greater than s .

11 Electrochemistry and Oxidation States

11.1 Ctrl+F keywords: oxidation state, redox, galvanic cell, anode, cathode, Nernst, concentration cell, electrode potential, elektrokemi

Use for oxidation numbers, spontaneous galvanic cells, standard cell potentials, Nernst calculations, concentration cells, and electrochemical equilibrium.

11.2 Symbols and notation

Table 11.1: Symbols and notation for electrochemistry.

Symbol	Meaning and usage
$E_{\text{cell}}, E_{\text{cell}}^\circ$	Cell potential and standard cell potential.
x	Unknown oxidation state in an oxidation-number calculation.
$E_{\text{cathode}}^\circ, E_{\text{anode}}^\circ$	Tabulated standard reduction potentials assigned to the cathode and anode.
$\Delta G^\circ, n, F, R, T$	Standard Gibbs energy change, transferred electrons, Faraday constant, gas constant, and absolute temperature.
K, Q	Equilibrium constant and reaction quotient for the balanced cell reaction.
$c_{\text{high}}, c_{\text{low}}$	Higher and lower metal-ion concentrations in a concentration cell.
,	Phase boundary and salt bridge in cell notation.

11.3 Oxidation states

11.3.1 Oxidation state rules

Table 11.2: Fast oxidation-state assignment rules.

Species or condition	Oxidation state or sum
free element	0
monatomic ion	ion charge
oxygen in most compounds	$-II$
hydrogen in most compounds	$+I$
halogens in binary salts	$-I$
neutral molecule	sum = 0
polyatomic ion	sum = ion charge
element whose oxidation state increases is oxidized.	

Oxidation states are bookkeeping numbers for redox. The

11.4 Exam recipe: oxidation number

Use when: an oxidation state of Cr, S, Mn, N, or another atom is requested in a compound or ion.

Given / Find: Given a molecular or ionic formula and its overall charge; find the oxidation state x of the specified element.

Required references: Table 11.2.

Steps:

1. Assign the fixed oxidation states from Table 11.2, including their atom multiplicities.
2. Set the requested atom's oxidation state to x , multiplied by its subscript if more than one occurs.
3. Set the sum of all oxidation-state contributions equal to the overall species charge.
4. Solve for x and state whether it indicates oxidation or reduction only if a comparison state is supplied.

Checks: Reinsert x and recover the overall charge; free elements such as O_2 , H_2 , and elemental metals have oxidation state 0.

Common traps: Setting an ion sum to zero; ignoring subscripts; applying the usual oxygen or hydrogen rule without checking for an indicated exception.

11.5 Galvanic cells

11.5.1 Anode, cathode, and electron flow

$$\text{anode: oxidation,} \quad \text{cathode: reduction,} \quad e^- : \text{anode} \rightarrow \text{cathode} \quad (11.1)$$

In a spontaneous galvanic cell, the cathode has the more positive reduction potential.

11.5.2 Cell notation



Single bars are phase boundaries and the double bar is the salt bridge. Anode is written on the left by convention.

11.5.3 Standard cell potential

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} \quad (11.3)$$

Use tabulated values as reduction potentials. Do not multiply E° by stoichiometric coefficients.

11.5.4 Gibbs free energy and equilibrium

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ} \quad (11.4)$$

$$nFE_{\text{cell}}^{\circ} = RT \ln K \quad (11.5)$$

Positive E_{cell}° gives negative ΔG° and product-favored equilibrium.

11.6 Exam recipe: galvanic cell potential and spontaneous direction

Use when: two half-cell reduction potentials are given or anode, cathode, electron flow, E_{cell}° , ΔG° , or K is requested.

Given / Find: Given two standard reduction half-reactions and potentials; find spontaneous direction and the requested cell quantity.

Required references: Equations (11.1), (11.3), (11.4), and (11.5).

Python: Brug `exam_toolbox.ipynb` med `equilibrium_constant(...)` efter $\Delta G^{\circ} = -nFE^{\circ}$ er beregnet, eller `constant("F")` ved manuel ΔG° -beregning.

Steps:

1. Select the more positive reduction potential as reduction at the cathode.
2. Reverse the other listed half-reaction for oxidation at the anode and balance transferred electrons to determine n .
3. Compute E_{cell}° with Equation (11.3); keep tabulated potentials unscaled.

4. Write electron flow from anode to cathode and, if needed, cell notation with the anode on the left.
5. Use n with Equation (11.4) or (11.5) only when ΔG° or K is required.

Checks: A spontaneously written galvanic cell has $E_{\text{cell}}^\circ > 0$ and $\Delta G^\circ < 0$; the half-reactions must cancel electrons.

Common traps: Multiplying potentials by stoichiometric coefficients; reversing the anode reaction without using its reduction potential consistently in the subtraction; sending electrons from cathode to anode.

11.7 Nernst equation and concentration cells

11.7.1 Nernst equation

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q \quad (11.6)$$

The reaction quotient is written for the balanced cell reaction.

11.7.2 Nernst equation at 298 K

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0592 \text{ V}}{n} \log Q \quad (11.7)$$

Use base-10 logarithm only with 0.0592 V. Use n as moles of electrons transferred.

11.7.3 Concentration cell

$$E_{\text{cell}}^\circ = 0 \quad (11.8)$$

$$E_{\text{cell}} = \frac{0.0592 \text{ V}}{n} \log \left(\frac{c_{\text{high}}}{c_{\text{low}}} \right) \quad (11.9)$$

For a metal/metal-ion concentration cell written spontaneously, the high-concentration half-cell is reduced and the low-concentration half-cell is oxidized.

11.7.4 Solubility product from cell potential

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ = -RT \ln K \quad (11.10)$$

Write the cell reaction so its equilibrium constant matches the desired dissolution or precipitation expression.

11.8 Exam recipe: solubility product from electrochemical potential

Use when: standard reduction potentials or a standard cell potential are given and the requested equilibrium constant is K_{sp} for a sparingly soluble salt.

Given / Find: Given relevant half-reaction potentials and salt dissolution species; find K_{sp} .

Required references: Equations (11.3), (11.10), and (10.14).

Python: Brug `exam_toolbox.ipynb` med `equilibrium_constant(...)` efter $E^\circ \rightarrow \Delta G^\circ$; brug `constant("F")` og `constant("R")` til manuel kontrol.

Steps:

1. Write and balance half-reactions so their sum is the salt dissolution reaction whose equilibrium expression is K_{sp} , rather than its reverse precipitation reaction.
2. Determine the electron number n , and calculate E_{cell}° for that oriented overall reaction using Equation (11.3); reverse the sign if the summed reaction was first obtained backwards.
3. Calculate $\ln K_{sp} = nFE_{\text{cell}}^\circ/(RT)$ from Equation (11.10), then exponentiate to obtain K_{sp} .
4. Check that the resulting equilibrium expression has the dissolved ions with the powers required by Equation (10.14).

Checks: A sparingly soluble salt normally gives $K_{sp} \ll 1$, corresponding to negative E° for dissolution as written; the balanced dissolution reaction contains no free electrons.

Common traps: reporting $1/K_{sp}$ by orienting the reaction as precipitation; multiplying an electrode potential by coefficients; using the wrong electron count in the logarithmic relation.

11.9 Exam recipe: Nernst concentration cell

Use when: identical electrodes contact different ion concentrations, for example copper or nickel concentration cells.

Given / Find: Given two concentrations, temperature 298 K, and the electrode ion charge; find cell voltage and spontaneous electrode direction.

Required references: Equations (11.8) and (11.9); use Equation (11.6) instead if the temperature is not 298 K.

Python: Brug `exam_toolbox.ipynb` med `cell_potential(...)` for Nernst-beregninger og koncentrationsceller, især naar $T \neq 298$ K.

Steps:

1. Confirm that the electrodes and redox couple are identical, so $E_{\text{cell}}^{\circ} = 0$.
2. Determine n from the balanced metal oxidation/reduction half-reaction.
3. Assign c_{high} and c_{low} , and write the spontaneous direction as reduction in the more concentrated metal-ion half-cell.
4. Insert the ratio $c_{\text{high}}/c_{\text{low}}$ into Equation (11.9).
5. Report the positive cell potential for the spontaneous orientation and identify anode and cathode when requested.

Checks: The voltage is positive for the spontaneous orientation and becomes zero when the concentrations are equal; the logarithm argument is at least one.

Common traps: Inverting the concentration ratio; using $n = 1$ without checking ion charge; adding a nonzero standard potential for identical half-cells.

12 Exam Strategy and Mixed Task Recipes

12.1 Ctrl+F keywords: multiple choice, statement validity, units, limiting case, answer check, traps, Danish keywords

Use for cross-topic exam tactics that recur in marked multiple-choice and calculation problems.

12.2 Symbols and notation

Table 12.1: Symbols used in cross-topic checking strategies.

Symbol	Meaning and usage
pH, $[\text{H}_3\text{O}^+]$	pH and hydronium concentration used in logarithmic answer checks.
ΔG° , K	Standard Gibbs energy change and equilibrium constant.
E , E° , n , Q	Cell potential, standard potential, transferred-electron amount, and reaction quotient in Nernst checks.
n_{gas} , p , V , R , T	Gas amount, pressure, volume, gas constant, and absolute temperature.

12.3 Exam recipe: multiple-choice statement elimination

12.3.1 One false clause makes an option false

$$\text{valid answer option} = \text{all required clauses true} \quad (12.1)$$

For statement options, test each clause independently: sign, unit, condition, and exception. Eliminate options with one wrong condition even if the formula name looks right.

12.3.2 Dimensional and unit check

$$\text{final unit must match requested quantity} \quad (12.2)$$

Use units as algebraic factors. Common exam errors are Celsius in gas/Arrhenius equations, entropy in J with enthalpy in kJ, and cell length units in density.

12.3.3 Logarithm checks for pH, K , and Nernst

$$\text{pH} = -\log[\text{H}_3\text{O}^+], \quad \Delta G^\circ = -RT \ln K, \quad E = E^\circ - \frac{0.0592}{n} \log Q \quad (12.3)$$

A factor 10 in concentration changes pH or Nernst terms by one logarithmic unit. Check whether ln or log belongs with the numerical constant.

Use when: the question asks which statement or numerical option is correct, or several options differ only in sign, unit, condition, or direction.

Given / Find: one or more candidate answers; determine which option satisfies every required condition.

Required references: Equation (12.1), Equation (12.2), Equation (12.3), and the numbered topic equations or tables named by each option.

Steps:

1. Split each answer option into separate claims, for example formula choice, sign, unit, phase inclusion, direction, or stated assumption.
2. For each formula claim, locate the matching numbered topic relation and substitute or compare exactly as that relation states; do not accept an option because its topic name sounds plausible.
3. For each numerical claim, carry units through the calculation and eliminate the option if its final unit fails Equation (12.2).
4. For pH, equilibrium, or electrochemical logarithm claims, verify logarithm base and sign using Equation (12.3).
5. Eliminate an option immediately when any single claim is false, according to Equation (12.1); retain only options whose claims all pass.

Checks: Re-read qualifiers such as “standard”, “at equilibrium”, “spontaneous”, “largest”, or “lowest” before submitting the retained option.

Common traps: checking only the first clause of a compound option; accepting a correct formula with wrong units or wrong logarithm sign; overlooking a stated condition.

12.4 Recurring mixed recipes

12.4.1 Ideal gas stoichiometry

$$n_{\text{gas}} = \frac{pV}{RT} \quad (12.4)$$

Convert gas data to moles before applying reaction coefficients. This links gas-law questions with limiting-reactant calculations.

12.4.2 Percent change in hydronium concentration from pH

$$\frac{[\text{H}_3\text{O}^+]_2}{[\text{H}_3\text{O}^+]_1} = 10^{\text{pH}_1 - \text{pH}_2}, \quad \% \text{ change} = (\text{ratio} - 1)100\% \quad (12.5)$$

Use for ocean-acidification or pH-change wording. Lower pH means larger hydronium concentration.

12.4.3 Recognition keywords, English and Danish

Table 12.2: Recognition keywords and the topic recipe they identify.

Exam wording	Go to
bundfald, precipitate, sparingly soluble	K_{sp}, Q_{sp}
puffer, buffer, Henderson	acid/base buffer
galvanisk, electrode, concentration cell	Nernst/electrochemistry
planar, bond angle, pyramidal	Lewis/VSEPR
limiting, excess, yield, udbytte	stoichiometry
vapor pressure, boiling point	intermolecular forces
rate law, half-life, Arrhenius	kinetics

Use the wording to jump to the relevant recipe before doing algebra.